

Program overview

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Year 2022/2023
 Organization Industrial Design Engineering
 Education Bachelor Industrial Design Engineering

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Bachelor IO 2022			
1e jaar Bachelor IO	1st year Bachelor IO		
IOB1-1-22	DP1: Diving into Design	10	
IOB1-2	Understanding Product Engineering	5	
IOB1-3	Understanding Design	5	
IOB1-4	Understanding Organisations	5	
IOB1-5	Understanding Humans	5	
IOB2-1-22	DP2: Designing Product Services	10	
IOB2-2	Digital Product Development	5	
IOB2-3	Research for Design	5	
IOB2-4	Understanding Values	5	
IOB2-5	Digital Interfaces	5	
2e jaar Bachelor IO	2nd year Bachelor IO		
IOB3-1-22	DP3: Designing System Interventions	10	
IOB3-2-22	Product Engineering	5	
IOB3-3-22	Data	5	
IOB3-4	Envisioning the Future	5	
IOB3-5	Sustainable Impact	5	
IOB4-1-22	DP4: Designing for Society	10	
Keuzevak Organisations	Elective Organisations		
IOB4-B1-22	Strategic Brand Management	5	
IOB4-B2	Business Model Design	5	
IOB4-B3-22	Codesign in Services	5	
Keuzevak Skills	Elective Skills		
IOB4-S1-22	Design Communication and Visualisation	5	
IOB4-S2-22	Leadership and Project Organisation	5	
IOB4-S3	Mastering Research Methods	5	
Keuzevak Technology	Elective Technology		
IOB4-T1-22	Materials and Manufacturing	5	
IOB4-T2	Mechatronics	5	
IOB4-T3	Machine Learning for Design	5	
Keuzevak People	Elective People		
IOB4-P1	Human Factors & Ergonomics	5	
IOB4-P2	Experience, Motivation & Behaviour	5	
IOB4-P3	Culture & Society	5	
3e jaar Bachelor IO	3rd year Bachelor IO		
IOB6-1-22	DP5: Bachelor Final Project	15	
Keuzevakken			
IOB6-E1	Design for the Circular Economy	5	
IOB6-E10	Design Didactics	5	
IOB6-E2	Reimagining Mobility	5	
IOB6-E3	Health	5	
IOB6-E4-22	Identity and Portfolio	5	
IOB6-E5	Entrepreneurial Innovation by Design	5	
IOB6-E6	Design Engineering Contest	5	
IOB6-E7	Biomechanics of Product Use	5	
IOB6-E8	Design Analytics	5	
IOB6-E9-22	Form and Senses	5	

Year	2022/2023
Organization	Industrial Design Engineering
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Bachelor IO 2022

Year	2022/2023
Organization	Industrial Design Engineering
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1e jaar Bachelor IO	
Program Coordinator	R.P. de Vos

IOB1-1-22	DP1: Diving into Design	10
Responsible Instructor	S.R. Dehli	
Contact Hours / Week x/x/x/x		
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	Dutch	
Course Contents	In het vak DP1 Diving into Design duiken studenten in het ontwerpen en gaan ze 3 ontwerpuitdagingen aan. Tijdens de duik krijgen studenten inzicht in de 6 hoofdcompetenties van het ontwerpen navigeren, framing, concepting, visualiseren & prototypen, evalueren en integreren - en zetten ze de eerste stappen in het aanleren van de vaardigheden die daaraan gekoppeld zijn. Zo beginnen ze aan de klim om vanuit een bewuste onbekwaamheid, bekwaam te worden in het ontwerpen. Daarnaast beginnen studenten in DP1 aan hun eigen professionele ontwikkeling (onder de noemer IDME) waarmee ze uiteindelijk in staat zijn om zich te ontwikkelen tot de soort ontwerper die ze willen worden. De eerste stap hierin en de focus voor deze ontwikkeling in DP1, is het leren studeren en begrijpen dat je als student zelf verantwoordelijk bent voor of en hoeveel je leert. De focus van DP1 ligt op het aanleren van het iteratieve ontwerpproces en de stappen van de ontwerpcyclus daarin. Zo gaan studenten in meerdere iteraties erop uit om te begrijpen waarvoor en wat ze moeten ontwerpen, hoe ze dat kunnen ontwerpen door veel ideeën te genereren en deze ideeën te schetsen, bouwen en 3D modelleren, en of dat wat ze ontwerpen in de praktijk doet wat het zou moeten doen. Hiervoor passen studenten aangereikte methodes toe en kennis uit de parallelle vakken Understanding Product Engineering, Understanding Humans en Understanding Organisations om een ontwerp te ontwikkelen waarin rekening is gehouden met zowel werking en constructie, functionaliteit en gebruik, als vorm en uitstraling. Het leren ontwerpen is hiervoor gericht op iets tastbaars wat redelijk dicht bij de belevingswereld van de studenten. Het uitgangspunt voor alle projecten is het ontwerpen van een fysiek product voor een gegeven organisatie en doelgroep.	
Study Goals	Als studenten het vak afgerond hebben, kunnen ze: 1. een eigen studeerstrategie formuleren, met behulp van feedback kritisch reflecteren op de uitvoering ervan en hun eigen handelen daarop aanpassen (5.1.1, 5.1.3, 5.4.3) 2. aantonen dat ze aangereikte methodes voor de verschillende stappen van de ontwerpcyclus in meerdere iteraties kunnen toepassen (1.1.2), 3. laten zien dat ze de context van een gegeven ontwerp kader vanuit de perspectieven mens, organisatie en technologie kunnen verkennen in een iteratief proces (1.3.1), 4. demonstreren dat ze vanuit een gegeven ontwerp kader een oplossingsruimte in de breedte kunnen verkennen (1.4.1), 5. door het maken van simpele schetsen, prototypes en 3D modellen demonstreren dat ze kunnen reflecteren op hun eigen ontwerp en het ontwerp verder kunnen ontwikkelen (1.5.1), 6. beoordelen of hun ontwerp voldoet aan het ontwerp kader door experimentele testen uit te voeren (1.6.1), 7. aantonen dat ze rekening houden met verschillende ontwerp aspecten (de werking en constructie, de functionaliteit en het gebruik, de vorm en uitstraling) in de conceptontwikkeling van een ontwerp (1.2.2).	
Education Method	De ontwerpprojecten: Project 1 is een mini-project van 2 weken, een soort kennismaking met de faculteit waarin studenten laten ze wat zij denken dat ontwerpen is. In dit project werken studenten in kleine teams. Project 2 en Project 3 zijn respectievelijk 6 en 8 weken lang en studenten doorlopen respectievelijk 3 en 4 iteraties van de ontwerpcyclus zodat ze de leerdoelen van het vak veel kunnen oefenen en zich eigen maken met ruimte voor vallen, opstaan en experimenteren. In deze projecten werken studenten individueel. In alle projecten werken studenten aan ontwerp opdrachten gefocust op de eerste kwadrant van de Navigation Canvas (van Erp) - het realiseren van een ontwerp vanuit een gegeven ontwerp kader. De werkvormen: · Colleges en gastlezingen · Workshops: Tijdens de ontwerpprojecten volgen studenten workshops onder leiding van een ontwerpdocent. Voor de skills visualiseren, prototyping en 3D modelleren volgen studenten aparte workshop reeksen onder leiding van experts. Voor het leren studeren (IDME) volgen studenten een reeks workshops onder leiding van een teaching assistant. · Coaching: Tijdens de ontwerpprojecten krijgen studenten coaching in kleine groepen van een ontwerpdocent. · Zelfstudie · Consults: Voor de skills visualiseren, prototyping en 3D modelleren is er bijna wekelijks 1 dagdeel beschikbaar voor feedback en begeleiding in het uitvoeren van de zelfstudie.	
Literature and Study Materials	Verplichte literatuur bij dit vak: - Delft Design Guide, van Boeijen, A., Daalhuizen, J., & Zijlstra, J. (2020). Delft Design Guide: Perspectives, models, approaches, methods. (2nd ed.) BIS Publishers.	
Prerequisites	Alleen studenten die zijn geselecteerd voor de bachelor Industrieel Ontwerpen aan de TU Delft, na het afronden van selectieprocedure, kunnen tot IO bachelor vakken worden toegelaten (artikel 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	In DP1 worden studenten op twee momenten summatief beoordeeld: - na Project 2 halverwege het vak, - na Project 3 aan het einde van het vak. Project 2 wordt beoordeeld met een pass of een voorlopige fail, met andere woorden een fail met de optie om aan te vullen. Het aanvullende werk wordt gedaan tijdens Project 3. Project 3 wordt beoordeeld met een cijfer en telt daarmee voor 100% in het cijfer van het vak als geheel. Als het cijfer voor Project 3 een 6,0 of hoger is, wordt een mogelijke voorlopige fail voor Project 2 aangepast naar een pass. Als het cijfer voor Project 3 een 5,5 of lager is, blijft een mogelijke voorlopige fail voor Project 2 staan. Leerdoelen 2-7 worden getoetst door eenzelfde deliverable voor beide toetsmomenten, een zogenaamde Design Evolution. Deze deliverable is een visuele weergave van de ontwikkeling van het ontwerp in het desbetreffende project met korte ondersteunende teksten. Het eerste toetsmoment dient daarom vooral voor het geven van feedback aan de student waar de student vervolgens mee aan de slag kan in Project 3. Leerdoel 1 wordt getoetst door een soortgelijk deliverable, een zogenaamde Student Evolution. Hierin laat de student zien hoe een persoonlijke studeerstrategie ontwikkeld en gevolgd wordt parallel aan het leren ontwerpen in Project 2. Met de studeerstrategie geven studenten aan en stellen ze zichzelf in staat om de andere 6 leerdoelen, met betrekking op het leren ontwerpen, te halen. Leerdoel 1 is onderdeel van de eerste summatieve toetsing halverwege het vak en maakt deel uit van de pass/fail beoordeling. Bij beide summatieve toetsmomenten van DP1 wordt je individueel beoordeeld. Tijdens het vak ontvangen studenten regelmatig formatieve feedback op zowel hun Design Evolutions als hun Student Evolution. Halverwege Project 3 wordt ook schriftelijke formatieve feedback gegeven.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.	
Special Information	Registration beyond the registration deadlines is not possible.	

IOB1-2	Understanding Product Engineering	5
Course Coordinator	Dr.ir. S.F.J. Flipsen	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course Contents	Om producten in de fysieke wereld te ontwerpen moet een ontwerper de technische basisprincipes kennen van het product en zijn onderdelen. Een gestructureerde technische analyse van bestaande producten is daarbij nodig om deze kennis beter te begrijpen en toe te passen in een toekomstige ontwerpcontext. Er bestaat een overweldigend aanbod aan materialen en productieprocessen, maar ook aan analysemogelijkheden om te voorspellen hoe producten (gaan) functioneren in verschillende situaties. Om wegwijs te worden in dit veld wordt de student geïntroduceerd in vijf technische aspecten van tastbare producten, namelijk: Functie, Productarchitectuur, Materialen, Productietechnieken en Modelleren, gerepresenteerd in het zogenaamde Techniekwiel. In dit vak staat het product centraal, van waarlangs het techniekwiel wordt doorlopen. Naast de functie van alle onderdelen in het product en de relatie tot elkaar, zal ieder onderdeel worden geanalyseerd op de meest voorkomende materialen en productietechnologieën. Daarnaast zal door middel van een systematische analyse van geometrie, uitwendige en inwendige belastingen, en de leer van spanningen en vervormingen de student inzicht krijgen in de technische haalbaarheid van onderdelen in de productarchitectuur. Dit vak legt de relatie uit tussen materiaalgroepen en hun belangrijkste eigenschappen, de belangrijkste verschillende productieprocessen die daarbij gebruikt worden, en de kansen en beperkingen die dit biedt voor productontwerp. De student maakt kennis met deze aspecten door het routineus verkennen van een tastbaar product, het te begrijpen door middel van analyse en abstractie, en het te evalueren op de vijf aspecten van het Techniekwiel.	
Study Goals	Als studenten het vak afgerond hebben, kunnen ze: 1. Semi-formele en formele methoden gebruiken om de productarchitectuur van producten te analyseren met betrekking tot technische en duurzaamheidsaspecten (2.5.1); 2. Analytische en experimentele methoden gebruiken om functionele en niet-functionele eigenschappen van een technische productarchitectuur te testen en te evalueren (2.5.2); 3. De basisprincipes van statica toepassen in productengineering o.a. vrije lichaamsdiagrammen en evenwichtsvergelijkingen (2.2.1); 4. Kennis toepassen over de meest voorkomende materialen en fabricageprocessen en de daarmee samenhangende mogelijkheden en beperkingen voor het ontwerp van producten (2.2.2); 5. De basisprincipes van de mechanica van materialen en materiaal kunde toepassen in de meest voorkomende constructiesituaties binnen de context van productontwerp (2.2.2); 6. De basis wiskunde toepassen in de context van productontwerp (2.1.1); 7. Een afgewogen oordeel maken over de materiaalkeuze voor het creëren van een ecologisch duurzaam product (3.3.2).	
Education Method	De kennis wordt overgedragen in hoorcolleges, welke vervolgens in actieve test en demonteer workshops wordt gebruikt voor de technische analyse van het product en zijn onderdelen. Het vak bestaat uit hoorcolleges (theorie en modellen), workshops (experimenteren en ervaren) en peer-feedbacksessies (observeren en reflecteren): Hoorcolleges: elke week begint met een college van twee uur. Hierin komt de technische analyse van de producten aan bod. Daarnaast wordt de student de kennis en vaardigheden aangeboden om producten en hun architectuur technisch te analyseren. Workshops (in groepen van 5): tijdens de wekelijkse workshops van 4 uur analyseren studenten de vijf aspecten van het techniekwiel middels een testopstelling, object, consumentenproduct, model of berekening. Deze analyses worden gebruikt om actief de stof van de week te ervaren en te begrijpen. Alle vijf basisaspecten komen aan bod. In de laatste weken van het vak worden naar aanleiding van de analyses voorstellen gegenereerd voor aanpassing op bepaalde aspecten, waarin de mogelijkheden van aanpassingen van materiaalgebruik, combineren van functies, andere gebruikssituaties, et cetera. Hieruit ontwikkelt de student kritische vragen gebaseerd op de theoretisch en praktische kennis. Per workshop verschillen de producten en de focus op aspecten. Peer-feedback (in groepen van 5): De studenten documenteren de technische productanalyse met behulp van het techniekwiel en bespreken onderling de oplossingen, vergelijkingen, vragen en onderbouwing.	
Literature and Study Materials	1. Syllabus Understanding Product Engineering via Webdu 2. Mike Ashby & Kara Johnson Materials and Design (3ed), Elsevier, 2014 via download of optioneel als print. 3. optioneel: Jan van de Craats & Rob Bosch, Basisboek Wiskunde (2ed), Pearson, 2018	
Prerequisites	Alleen studenten die zijn geselecteerd voor de bachelor Industrieel Ontwerpen aan de TU Delft, na het afronden van selectieprocedure, kunnen tot IO bachelor vakken worden toegelaten (artikel 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	De leerdoelen van de cursus worden beoordeeld door middel van een individueel schriftelijk tentamen. Dit bepaalt voor 100% het eindcijfer.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: upe-io@tudelft.nl	

IOB1-3	Understanding Design	5
Course Coordinator	Dr.ir. A.I. Keller	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	Design shapes the world, as much as the world co-shapes design. But what is design? In order to understand what constitutes Industrial Design Engineering, we need to find out what vocabulary, practices and methods are commonly used here in Delft and around the world. What shifts in the triangle of technology, society, and economy have co-created designs vocabulary, practices, and methods? How have shifts in our understanding of consumption shaped the field? How have shifts in production co-constituted what design is now? How to situate seemingly newfangled terms like complexity and systems thinking? Understanding the history of the present field of design future-proofs students: it will ready them for the next iteration of the field.	
Study Goals	Upon finishing this course you will be able to: 1. Recognize and explain through concrete examples how design, society and culture influence each other (2.8.1, 2.8.2) 2. Recognize and contextualize the field of designs vocabulary, practices, and methods in relevant historical developments (3.1.1) 3. Critically reflect on knowledge, information, and common sense, and towards the domains of design and society (5.1.1.)	
Education Method	This course consists of plenary lectures, reading and multimedia assignments, and seminars. Each week is dedicated to one topic, with short cycles consisting of: 1. Setting the stage by introducing the topic in a lecture 2. Self-study of mandatory and optional materials for each topic (a blend of texts, podcasts, documentaries). Students are stimulated to deal with the content actively, critically, and in-depth, in a reading reflection for each topic. In this reading reflection the required material can be enriched with optional other material and their own insights, resulting in a short reflection, containing two questions for the seminar. 3. Weekly seminars in which the students will use the reading reflections with their questions to explore the topic. Under supervision of a seminar coach the students are stimulated to take a position towards the content and value the logic and evidence for this position, as well as develop their analytical and dialogic skills.	
Literature and Study Materials	http://understandingdesign.net	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	Active participation in the Thursday seminars is key to achieving the learning objectives in this course. This is a space to think, experiment and try out your thoughts. Students are expected to submit reading reflection in a timely fashion and to participate actively during the seminar. In order to create a safe environment for learning and discussion, student participation or performance is not measured or graded. The learning objectives of the course are assessed through (1) an exam and (2) a final submitted portfolio of reading reflections with a reading guide and a visual. The exam (1), in week 10 will account for 33% of the final grade. In week 9 the students will deliver a reading guide bundled with their reading reflections. This reading guide will account for 67% of the grade. All the assessments are individual.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: understandingdesign-io@tudelft.nl	

IOB1-4	Understanding Organisations	5
Course Coordinator	Ir. S.S. Mulder	
Contact Hours / Week x/x/x/x	3	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In this introductory course students are asked to look at organisations from three different interests: 1. a technical interest with a lens focusing on efficiency/control and cause/effect, 2. an interpretative interest with a lens focusing on the life experience of people in a cultural and organisational context, 3. a critical interest with a lens focusing beyond the status quo. Students will learn about key scholars, typical persons for each of these interests. Students will apply their knowledge of interest one and two by reporting on contemporary organisations by paying attention to the shaping of organisations, setting innovation goals, organising innovation, business economics, brands and organisational types & structures. Students will work with the third interest by devising a design brief while paying attention to ethics.	
Study Goals	Upon finishing this course students are able to: 1. distinguish between different types of innovation and analyse how these are related to design within organisations (LO 2.9.3). 2. distinguish between different types of systems and explain different types of behaviour of these systems (LO 3.2.2). 3. identify the organisations capabilities, the brand values and network relations in the current context (LO 2.9.1). 4. explain a basic business case based on estimated revenues and costs for a new design of a product-service system (LO 2.10.5). 5. apply knowledge of the variety of business models to develop an inclusive and sustainable product-service system that supports global prosperity (LO 3.3.4). 6. apply the principles of effective learning to collaborate in a project (LO 6.2.1) 7. identify social issues in design and acknowledge the professional responsibilities of designers (LO 3.4.1). 8. analyse the ethical values of a design challenge and use these as inspiration for design brief development (LO 3.4.2). 9. morally explain design brief decisions and to discuss ethical issues in a design context (LO 3.4.3).	
Education Method	Students are approached as a knowledge producer in two forms: exemplary learning and collaborating in a research-like-way. What students learn should matter in the here and now but also invite them go beyond the status quo. Exemplary Learning Via exemplars, students progress from specific cases to more widely applicable knowledge within a studio (5 to 6 groups of 5 students) supported by staff. Exemplars are disclosed via a reader (literature excerpts illustrated with historic examples), via podcasts or a documentary. Students write short items both in small groups and individually and in small groups and share their produced knowledge through a design and business-related newspaper per studio. In week 6, before the Christmas break, students deliver their final newspaper for assessment. Collaborating in a research-like-way Via an individual assignment students convene with a researcher producing a short design brief that addresses a societal problem that both student and researcher find relevant. The researcher invites students at the beginning of the course where he/she explains their own research interest. Students apply to be matched to a researcher and form a class. At selected moments the class will elevate individual progress e.g. by way of a studio seminar. At the end of the course, in week 9, students submit a short design brief. This brief will not be assessed, but serves as a preparation for an exam in week 10 to test the understanding of business economics and the interplay between ethics, organisation, design and innovation.	
Literature and Study Materials	Mandatory literature & Study materials: - Reader IOB1-4 Understanding Organisations - Case materials will be provided via Brightspace & a website Recommended literature: Grey, C. (2021). A very short, fairly interesting and reasonably cheap book about studying organizations (5th ed.). London: SAGE.	
Reader	A course reader will be made available via EduWeb	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	Learning objectives 1 to 6 are covered by a group assignment (weeks 1-6) leading to a newspaper which will be assessed via a rubric. Each students individual contribution towards the group is assessed by a) individual items in the newspaper and authorship in group items b) evidence of development of applying the principles of effective learning to collaborate in a project This will form 70% of the final grade (grade minimum of 5,0). The group grade will be adjusted based on the individual contribution. Learning objectives 1 to 5 and 7 to 9 are covered by the individual design brief to prepare for the exam. This exam will form 30% of the final grade (minimum of 5,0). The exam consists of multiple-choice questions and open questions.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: uo-io@tudelft.nl	

IOB1-5	Understanding Humans	5
Course Coordinator	Dr. V.T. Visch	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Course Contents	In de cursus Understanding Humans staat de mens centraal en wordt hierover design-relevante kennis aangeboden. De cursus geeft een introductie in het waarom, hoe en wanneer mensen relevant zijn voor ontwerpers. Op deze basiskennis zullen de andere vakken en met name de people electives in Semester 4 voortborduren. We zullen 3 human kennis domeinen langsgaan: Human Factors (met fysieke-, perceptuele-, en cognitieve ergonomie); Human Experience (met emoties, esthetica, betekenis, motivatie en gedrag); Human Context (met culturele-, sociale-, politieke context, en de context van het dagelijkse leven).	
Study Goals	De Vitruvian Man zal het leidende karakter in het vak zijn. Hij zal zich steeds beter laten begrijpen en problematiseren. 2.6.1 Students are able to explain design-relevant factors of the human body and mind, including bodily functioning, the perceptual system, and information processing. 2.6.2 Students are able to explain design-relevant factors of human experience, including emotions, aesthetics, and meaning. 2.6.3 Students are able to explain design-relevant factors of human motivation and behaviour, focusing on the internal and external factors that influence human behaviour. 2.8.1 Students are able to explain design-relevant factors of cultural, sociological, political, economic and everyday-life contexts that stage and shape social interactions. 4.1.1 Students are able to assess the value of different types of research and their outcomes. 4.1.2 Students are able to assess the value of different sources of information/ knowledge. 5.1.1 Students are able to express critical reflect their own thinking, decision making and actions and adjust these accordingly, and reflect critically toward the domain. 5.2.1 Students are able to understand, justify, value, develop and explain theoretical arguments. 6.3.3 Students are able to present research and design outcomes clearly and convincingly in writing, verbally and visually.	
Education Method	De cursus bestaat uit wekelijkse cycli van colleges, eigen ervaringen, discussies, wetenschappelijke papers, essays en presentaties. Tijdens de colleges op maandag zal een team van facultaire onderzoekers een set van kernbegrippen behandelen en een wetenschappelijke paper introduceren. Hierna volgt een Weekly Wonder Assignment waarin de student de kernbegrippen zelf kan ervaren. Op donderdagen worden in een groep van 25 studenten deze ervaringen en presentaties besproken en gekoppeld aan de wetenschappelijk theorie uit de colleges en de paper. Vooraf aan de donderdag discussies wordt van iedere student verwacht dat de paper is gelezen en bijbehorende opdracht is gemaakt, de weekly wonder assignment is uitgevoerd en dat hij/zij een object/beeld/ muziekstuk upload dat refereert aan de weekly wonder assignment. Tijdens iedere donderdag discussie zullen steeds 3 wisselende studenten een korte presentatie geven: 2 over de weekly wonder assignment en 1 over de toepassing van de kernbegrippen van de week op het parallel lopende ontwerpvak. Na deze presentatie zal er een groepsdiscussie volgen volgens de Socratische discussie principes van Analyse, Toepassing en Evaluatie waarin a) gereflecteerd wordt op de ervaringen, b) de ervaringen worden verbonden met de theoretische kernbegrippen, c) de relatie met de theorie, ervaringen en design wordt besproken en d) een individuele kritische beargumenteerde positie tov de theorie wordt gevormd. Tijdens de discussie sessies worden kritisch denken, begrip van theorie en praktijk, en discussie/argumentatie vaardigheden getraind. Dit wordt getoetst door 3 essay die gedurende de cursus door iedere student geschreven wordt. De cursus sluit af met een toets over de kennis van de kernbegrippen.	
Literature and Study Materials	- wetenschappelijke papers - lecture slides - boekje met kernbegrippen - course manual	
Prerequisites	Alleen studenten die zijn geselecteerd voor de bachelor Industrieel Ontwerpen aan de TU Delft, na het afronden van selectieprocedure, kunnen tot IO bachelor vakken worden toegelaten (artikel 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	Understanding Humans zal zowel formatief als summatief getoetst worden. Formatieve toetsing: - De wekelijkse discussie sessies inclusief de voorbereidingen. - De presentatie tijdens de discussies. Ieder student zal 1 presentatie geven. - Feedback op essay #1 en #2 in week 3 en 6. Summatieve toetsing: - Gebundelde essays #1,2,3 in week 9, inclusief rapportage van voorbereidingen discussie sessies (Weekly Wonder Assignments, paper assignments) - Multiple choice test over de keyterms in week 10.	
Enrolment / Application	De essays tellen voor 70% mee in het eindcijfer, de keyterm test voor 30%. For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.	
Special Information	Registration beyond the registration deadlines is not possible. Contact: uh-io@tudelft.nl	

IOB2-1-22	DP2: Designing Product Services	10
Course Coordinator	Ir. C.P.J.M. Kroon	
Contact Hours / Week x/x/x/x		
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Course Contents	Digitale technologieën heeft een grote impact op nagenoeg alle sectoren van onze economie. Ze dragen bij aan de ontwikkeling van innovatieve ontwikkelingen, zoals nieuwe producten, platforms en services. Ook andere manieren van waarde ontwikkeling ontstaan, zoals nieuwe klant ervaringen (customer experiences). Organisaties doen er vaak goed aan om digitale technologieën in hun korte- en lange termijn strategieën op te nemen. Het doel van dit project-gebaseerde vak is om studenten lean innovatie methodes toe te laten passen voor een bestaande organisatie. Dit bedrijf heeft bepaalde zaken die ze goed doen. Vanuit hier gaan de studenten op zoek naar een manier om nieuwe impact te maken in de levens van mensen. Niet alleen voor de klanten, maar voor de verschillende stakeholders die betrokken zijn bij deze waardecreatie. Dit vak is ook een platform voor studenten om eerder opgedane (theoretische) kennis toe te passen, zoals hun kennis over design research, digitale producten, interaction design en het begrijpen van wat waarde is voor verschillende stakeholders. Het vak bestaat uit 3 fases. 1. In het eerste deel gaan studenten individueel de behoeftes van de gebruikers onderzoeken in de context van waar het bedrijf actief is. De uitkomst zal uiteindelijk een concept product-service systeem zijn. 2. In het tweede deel gaan studenten in teams van vijf één concept doorontwikkelen middels de lean startup methode. Ze zullen in 3 sprints aannames op desirability, viability en feasibility proberen te valideren. 3. In het derde en laatste deel gaat de student individueel de waarde van het uiteindelijke concept onderzoeken. Hierin analyseert hij/zij de waarde voor verschillende stakeholders en werkt zij/hij een passend, duurzaam business model uit.	
Study Goals	Als studenten dit vak hebben afgerond, zijn ze in staat om: LO1: een project te plannen en om een agile innovatie proces te managen om een mvp te maken (1.1.3) LO2: verschillende belangen van de stakeholders af te wegen om een haalbaar, wenselijk en levensvatbaar ontwerp met een optimale impact te creëren (1.2.2) LO3: het ontastbaar te laten zien en buiten de gebaande paden te treden ten behoeve van innovatie (1.4.3) LO4: visualisaties, modellen en prototypes te maken voor het effectief beoordelen van een concept (1.5.3) LO5: experimenteel te testen welke ontworpen functies/uitkomsten van het concept het bedoelde effect hebben (1.6.1) LO6: de belangrijkste trends op het gebied van technologie, organisatie en mensen te analyseren en dit toe te passen in hun concept (3.1.3) LO7: om de uitkomst van onderzoeken en experimenten te gebruiken tijdens het ontwerp proces (4.3.3) LO8: onderzoek, experiment en design output op een duidelijk, onderbouwde manier te communiceren, zowel in schrift, verbaal als visueel (6.3.2)	
Education Method	In dit vak bootsen we na hoe lean innovation methodes worden toegepast in de praktijk. Hierin zoomen we in op welke manier deze bijdragen aan de desirability, viability en feasibility van producten en services. Deze methodes worden uitgelegd middels blended learning, colleges, gastlezingen van bedrijven, workshops en coaching door experts. Een aantal lezingen geven de inhoud op het gebied van agile design methodes en de toepassing in de praktijk, dit varieert van eens per week tot 2 keer per maand. Iedere ±2 weken zijn er workshops in groepjes van vijf. Hierin geven we de studenten de mogelijkheid om vanuit de theorie hun experimenten voor te bereiden, maar ook het bediscussieren van hun assumpties en planning van de experimenten. Het groepswork van het project zal in groepjes van vijf gebeuren. Hierin focussen we op het snel ontwikkelen van concepten en het prototypen van (delen van) het digitale product/service concept, waar je je programmeervaardigheden ontwikkelt. Deze vaardigheden helpen je een low fidelity prototype te maken zodat je iteratief de feasibility, desirability en wellicht ook viability van je concept kan toetsen. Ook de digitale ontwerp en interactie ontwerp skills komen hier aan bod. In dit vak zoomen we ook in op visualisatie vaardigheden. Hoe communiceer je je concept duidelijk? Deze kan je hier oefenen en worden beoordeeld.	
Literature and Study Materials	TBD	
Prerequisites	Alleen studenten die zijn geselecteerd voor de bachelor Industrieel Ontwerpen aan de TU Delft, na het afronden van selectieprocedure, kunnen tot IO bachelor vakken worden toegelaten (artikel 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	De leerdoelen worden beoordeeld door verschillende deliverables die door de fases van dit project gaan: 1) 1e concept: Een poster van het concept met use case scenario service blueprint (individueel, summatief 20% van het eindcijfer) 2) 2nd concept (Desirability): pitch deck (10-min PowerPoint presentation in pre-defined format), sprint overview A3, low fidelity prototype van het voorgestelde concept (Group/Formatief) 3) 3e concept (Viability): Verbeterd concept van het product-service system met een focus op viability, geupdatet pitch-deck, 5 min pitch naar investors (pre-def format), (Group/Formatief) 4) 4e concept (feasibility): Verbeterd concept van het product service system (desirability, viability, feasibility). Final group pitch-deck (powerpoint in pre-def format), digital prototype, 10 min presentation (Group/Summatief, 55% van het eindcijfer) 5) Final concept: Kort verslag (3-5 pages), strategische fit, waarde en het future scenario (Individual/Summatief, 25% van het eindcijfer).	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.	
Special Information	Registration beyond the registration deadlines is not possible. Contact: dp2-io@tudelft.nl	

IOB2-2	Digital Product Development	5
Course Coordinator	J. Bourgeois	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides students with the technical foundation to design and develop innovative digital products. Students develop the ability to specify technical requirements such as Internet Architectures, database models and software behavior. They leverage this understanding to apply the business model canvas, select the appropriate development methods and reflect on their digital responsibility as designer. Throughout the course, students navigate the seven dimensions of the product canvas. It is the compass to grasp the dynamics of digital products and acquire the ability to effectively converse in a design and development process to realize digital products that are technically feasible and commercially viable. Students develop a hands-on understanding of core digital technologies and software development methods through Python programming, and apply techniques and practices for iteratively creating, validating and improving digital products.	
Study Goals	Upon finishing this course students are able to: 1. explain core concepts related to the realization of digital products, such as technical architecture, networks, databases, software development methods, business models, web technology and digital responsibility (2.5.2, 2.4.1, 2.4.3, 2.10.2) 2. specify algorithms and data structures driving the behavior of a digital product through pseudo code and UML diagrams (2.5.2) 3. read and write basic Python code to prototype a digital product (2.4.1) 4. analyze and evaluate the technical aspects (code, data, network) and business aspects (canvas) of a connected product (2.4.3, 2.10.2)	
Education Method	The course consists of 8 modules, a weekly cycle mixing theory and practice. There is a mix of online and offline activities. Each week involves: - the introduction of theoretical concepts (4hrs) through self-study video and reading materials; - live sessions (3hrs) including a programming sandpit and a panel of lecturers walking through some of the concepts in practice and offering students a key moment for questions; - a programming assignment (4hrs) to individually practice coding in self-content, digital product context, with gradual complexity - a group exercise (3hrs) to experience key concepts covered during the course.	
Literature and Study Materials	In each module, students read a book chapter accessible through the TU Delft library.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	Students will receive a formative assessment of LO1 through the crowd-sourced quizzes and Discourse interaction. They will receive formative assessment of LO2 and LO4 through the drop-in videos, structured peer feedback and live feedback session. They will receive a formative assessment of LO3 through example of solution code and Discourse interaction. Students will receive a summative assessment through an individual, open-book, online exam combining Multiple Choice Questions (LO1 and LO4), auto graded pieces of code (LO3) and open questions (LO2 and LO4). This exam accounts for 100% of the grade.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: dpd-io@tudelft.nl	

IOB2-3	Research for Design	5
Course Coordinator	Dr.ir. M. Melles	
Contact Hours / Week x/x/x/x	3	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	In Research for Design leren studenten over de waarde van onderzoek en wetenschappelijke kennis voor het ontwerpproces. Ze leren hoe ze bestaande kennis kunnen toepassen in het ontwerpproces en hoe ze zelf onderzoek kunnen opstellen en uitvoeren om nieuwe kennis te verkrijgen voor het ontwerpproces. Meer specifiek leren studenten hoe de verschillende fasen in het ontwerpproces vragen om verschillende onderzoeksbenaderingen. Studenten leren hoe ze onderzoeksvragen moeten opstellen, geschikte onderzoeksmethoden kunnen selecteren, gegevens kunnen verzamelen, analyseren, interpreteren en presenteren en, op basis van de resultaten, implicaties voor het ontwerp en het ontwerpproces kunnen definiëren. Tijdens Research for Design maken de studenten ook kennis met de verschillende onderzoeksthema's op de faculteit, door middel van de opdrachten, voorbeelden en gastsprekers. Een gedeelte van de geleerde onderzoeksvaardigheden wordt toegepast in Design Project 2 dat parallel aan dit vak loopt. Na dit vak zijn studenten in staat belangrijke onderzoekstermen te begrijpen, zelf ontwerp-gericht onderzoek op te stellen en uit te voeren en onderzoeksresultaten te integreren in het ontwerpproces. Studenten leren tevens om rekening te houden met ethische overwegingen bij het uitvoeren van onderzoek.	
Study Goals	Als studenten het vak hebben afgerond kunnen ze (getal tussen haakjes verwijst naar het leerdoel van de complete lijst eindtermen, waarvan het afgeleid is): LO1: de kwaliteit van verschillende soorten onderzoek en de resultaten ervan beoordelen (4.1.1) LO2: de kwaliteit van verschillende informatiebronnen en kennis beoordelen (4.1.2) LO3: adequate onderzoeksvragen formuleren (4.2.1) LO4: passende onderzoeksmethoden selecteren en toepassen om de onderzoeksvragen te beantwoorden (4.2.2) LO5: zowel kwantitatieve als kwalitatieve onderzoeksgegevens verwerken en analyseren, en hieruit conclusies trekken voor het ontwerp (4.2.3) LO6: rekening houden met ethische overwegingen van onderzoek (4.2.4) LO7: onderzoeksresultaten evalueren en reflecteren op generaliseerbaarheid en ontwerprelevantie (4.3.1) LO8: onderzoeksresultaten adequaat presenteren, rapporteren (in tekst en beeld) en bediscussiëren (4.3.2) LO9: onderzoeksresultaten toepassen tijdens het ontwerpproces (4.3.3) LO10: kritisch reflecteren op hun eigen denken, besluitvorming en handelen, en deze aan te passen (5.1.1) LO11: relevante argumenten voor hun ontwerpbeslissingen ontwikkelen en overtuigend presenteren (5.2.2)	
Education Method	Research for Design biedt wekelijks plenaire colleges om de theorie te bespreken en de opdrachten in te leiden, wekelijkse groepsopdrachten om de theorie toe te passen en onderzoeksvaardigheden op te doen en een contactmoment met de coach voor feedback op de weekopdracht. Elke week is er een plenair college waarin de belangrijkste onderwerpen van die week worden ingeleid en geïllustreerd. Van studenten wordt verwacht dat ze zich op deze interactieve colleges voorbereiden door literatuur te lezen en video's te bekijken (blended learning). Colleges worden verzorgd door experts van IO en gasten. Het hoofdonderwerp per week zal een specifieke onderzoeksmethode zijn gekoppeld aan het onderzoeken van de ontwerpprobleemruimte of de ontwerp oplossingsruimte. Algemene onderzoekskennis zoals het opstellen van onderzoeksvragen, het kiezen van de juiste methode, data-analyse en ethische overwegingen komen aan bod (en worden herhaald) gedurende het gehele vak. Studenten werken in een groep aan wekelijkse groepsopdrachten om hun onderzoeksvaardigheden te oefenen en de uitvoering en waarde van verschillende onderzoeksmethoden te ervaren. De groepen (vier studenten per groep) worden begeleid door een coach. Iedere opdracht is gericht op een specifieke onderzoeksmethode en wordt gestructureerd en gerapporteerd volgens een voorgeschreven template. Deze onderzoekstemplates zijn generiek en ook te gebruiken in andere vakken, zoals Design Project 2.	
Literature and Study Materials	Muratovski, G. (2021). Research for Designers (2nd Edition). Sage Publications, Ltd. (UK).	
Prerequisites	Alleen studenten die zijn geselecteerd voor de bachelor Industrieel Ontwerpen aan de TU Delft, na het afronden van selectieprocedure, kunnen tot IO bachelor vakken worden toegelaten (artikel 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	Het behalen van de leerdoelen wordt beoordeeld op basis van een groepsverslag en een individueel schriftelijk tentamen. Het groepsverslag rapporteert de opzet en uitkomsten van de wekelijkse opdrachten en is een bundeling van de weekopdrachten. Het is geschreven in wetenschappelijke stijl en gestructureerd volgens de voorgeschreven templates. Op basis van het verslag wordt toepassing van de theorie beoordeeld. Beoordeling van het verslag vindt plaats door middel van een rubric. Het individueel schriftelijk tentamen beoordeelt de kennis van de theorie en bestaat uit meerkeuze- en open vragen. Het eindcijfer voor Research for Design wordt berekend op basis van de deeltijfers voor het groepsverslag (60%) en het individuele schriftelijk tentamen (40%).	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: rfd-io@tudelft.nl	

IOB2-4	Understanding Values	5
Course Coordinator	Dr. L.B.M. Magnier	
Contact Hours / Week x/x/x/x	3-4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Understanding value(s) is a mandatory course which provides bachelor students with fundamental knowledge about the notions of value(s) and what they mean for a broad range of stakeholders. They will learn to identify the values that are important for all stakeholders involved in a new product-service system. They will learn to map-out these values, but also to analyse and explain possible tensions. Specifically, students will learn to understand how a value proposition can create value for the intended users, and how the organisations involved in putting this design on the market can capture economic value or revenue and therefore be viable. Importantly, students will learn to look at the consequences and externalities of a design. Societal and ecological value(s) will be scrutinised with a view to minimise harm and possibly increase benefits for indirectly affected stakeholders. In order to attend to value tensions emerging in design projects, students will get knowledge to better reflect on design decisions and outcomes using different ethical lenses. Eventually, students will be able to explain value flows in complex systems and to make a more desirable value proposition for all stakeholders. Through theoretical lectures and application in cases from practice in group projects, students will each week acquire knowledge, methods and skills that will enable them to incorporate ethics and stakeholder value(s) into their design process. The knowledge acquired in Understanding Values and the individual code of ethics that students will craft during the course will be relevant for all design projects in following courses as well as in a design career.	
Study Goals	Upon finishing this course students are able to: 1 - Students are able to distinguish between different notions of value 2 - Students are able to set up a value proposition for an intended user (2.10.3) 3 - Students are able to acknowledge the effect of design on human behaviour and reflect on how a design outcome promotes or hinders human values (2.6.3) 4 - Students are able to identify how an organisation intends to create and capture value (2.10.2) 5 - Students are able to identify relevant stakeholders for a new design and map the value flows to show what the value for each stakeholder is (2.10.1) 6 - Students are able to identify social and ecological issues in design and acknowledge the professional responsibilities of designers by assessing the value captured, missed and destroyed for both direct and indirect stakeholders (3.4.1) 7 - Students are able to ethically reflect on design challenges (3.4.2) 8 - Students are able to morally justify design decisions and to discuss ethical issues in different settings (3.4.3)	
Education Method	The course consists of lectures provided in different formats, coaching sessions and a project assignment. The lectures are held once a week on Monday morning and provide the student with theories and tools that will enable them to execute the project assignment, also some guest speakers will be invited. The coaching sessions are held every week on Wednesday afternoon and give students the opportunity to sit together and practice the theory with the support of their coach. They perform dialogues and get feedback on their progress. The project assignment is done in groups of four and consists of weekly assignments which collectively represent the final group report. The goal of these assignments is to apply the concepts, theories and tools provided during the lectures. Students will perform a thorough analysis of a PSS and eventually re-define its value proposition based on their conclusions. This value proposition will be presented together with a thoroughly argued value map and ethical reflection. Students work on their assignments during the coaching sessions and during self-study time.	
Literature and Study Materials	Literature and study materials will be provided via Brightspace	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of the course are assessed through a group report and an exam. The group report will be 60% of the grade. The individual exam will consist of multiple choice and will be 40% of the grade.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: uv-io@tudelft.nl	

IOB2-5	Digital Interfaces	5
Course Coordinator	Ing. A.J.C. van der Helm	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	Om interactieve producten te ontwerpen moet een ontwerper de basisprincipes van mens-machine interactie kennen. Deze principes worden verzameld uit drie vakgebieden: (1) de psychologie voor het begrijpen of bedenken van de gebruikers ervaring, (2) het ontwerpen voor vormgeving van interface en dynamisch product gedrag en (3) de electronica voor het kennen van componenten en programmeren van een computer om deze componenten te laten samenwerken in de beoogde functie van het product. Voor kennis over het begrijpen en bedenken van de gebruikers ervaring wordt voortgebouwd op het vak Understanding Humans. In Digital Interfaces wordt die kennis verdiept voor toepassing bij mens-machine interactie. Het ontwerpen van mens-machine interactie is als het ontwerpen van een theater-stuk met verschillende scenes waarbij de ontwerper vanuit de eerste persoon alle stappen van de gebruiker en de machine moet plannen, uitdenken en in de tijd achter elkaar zetten. Het omhelst zowel de vormgeving van de productvorm als de choreografie van het dynamisch gedrag. Voor kennis over de interactieve technologie wordt voortgebouwd op het vak Understanding Product Engineering en Digital Product Design. In Digital Interfaces wordt de anatomie van een product-service system behandeld waar het gaat om het meten van gebruiker acties (met sensoren), het schrijven van de software om de metingen om te zetten in een gewenste product reactie (met actuatoren).	
Study Goals	Als studenten het vak afgerond hebben, zijn ze in staat om: LO1 principes van menselijke ervaring toe te passen, waaronder multi-sensorische esthetische ervaringen, zingeving en emoties om een product-service systeem te ontwikkelen (2.6.2). LO2 principes van vorm, inclusief vormeigenschappen en ervaringsaspecten, toe te passen om een product-service systeem te ontwikkelen (2.7.2). LO3 de principes van dynamische gedragskwaliteiten, fysieke betrokkenheid en zingeving toe te passen om een product-service systeem te ontwikkelen (2.7.1). LO4 kennis over sensorische waarneming toe te passen voor het ontwikkelen van een interactief product-service systeem dat de omgeving en de mens waarneemt en daarop feedback geeft (2.4.2). LO5 kennis van interactieve technologie, inclusief elektronische basiscomponenten, sensoren en actuatoren, en ingebedde systemen toe te passen in de verwezenlijking van een ontwerp (2.3.1).	
Education Method	Het vak maakt gebruik van de volgende leer activiteiten: o) Een plenair college aan het begin van het vak o) Online video colleges met relevante theorie over mens-machine interactie o) Ieder video college heeft een quiz om opgenomen kennis te testen o) Activiteiten voor het maken en ervaren van interactieve prototypen o) Activiteiten voor het evalueren en reflecteren op het ontwerp en de werking van interactive prototypen o) Een individueel schriftelijk examen aan het einde van het vak Studenten werken individueel in het vak. In de eerste week van het vak wordt het mogelijk gemaakt een eigen pad te kiezen door het gebied van mens-machine interactie. Een pad dat de student op iedere stap een specifieke mens-machine interface laat ontdekken. Elke stap duurt één week en bestaat uit het individueel maken van een prototype van een interface en het gezamenlijk evalueren en reflecteren op gemaakte interfaces. Gedurende het vak kan voortschrijdend inzicht gebruikt worden om dit initiële pad aan te passen. Het maken van een interface wordt ondersteund door een tutorial dat de student kan uitvoeren als een recept. Er is een verzameling tutorials beschikbaar. Ieder tutorial heeft flexibiliteit in de keuze van sensoren, actuatoren, vormgeving etc. Gedurende de maak activiteiten is ondersteuning aanwezig. De programmeertaal is Python, hierin bouwt dit vak door op kennis ontwikkelt in Digital Product Design. De technische materialen voor het maken van de prototypen is beschikbaar in de IDE Electronics kit die studenten kunnen kopen op de faculteit. Het evalueren en reflecteren vindt iedere week plaats (na de eerste week) in peer-feedback sessies met andere studenten van het vak, deze sessies worden gefaciliteerd door experts. Voor het evalueren en reflecteren bouwt dit vak voort op kennis en routine ontwikkeld in het vak Research through Design. Door middel van de wekelijkse maak en peer-feedback sessies is een student in staat om mens-machine interactie te ontdekken op een persoonlijke manier, met aandacht voor persoonlijke voorkeuren voor specifieke vormen van mens-machine interactie. Door kennis te nemen van de prototypen van interfaces gemaakt door andere studenten ontstaat een rijk beeld van het vakgebied en de mogelijkheden voor mens-machine interactie. Het vakgebied is dusdanig groot dat het onmogelijk is om het compleet te behandelen. De verwachting is dat de student kennis verdiept in de design projecten en opvolgende vakken zoals Product Dynamics en electives zoals Mechatronics en Design Analytics.	
Literature and Study Materials	Het vak heeft een reader als aanvulling op de theorie behandeld in de online video colleges. De verzameling tutorials is beschikbaar online.	
Prerequisites	Alleen studenten die zijn geselecteerd voor de bachelor Industrieel Ontwerpen aan de TU Delft, na het afronden van selectieprocedure, kunnen tot IO bachelor vakken worden toegelaten (artikel 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	Tijdens een peer-feedback sessie worden de leerdoelen beoordeeld op een formatieve manier door middel van feedback van mede studenten. Het behalen van de leerdoelen wordt beoordeeld op basis van een individueel schriftelijk examen aan het einde van het vak en op basis van de maakactiviteiten. Het eindcijfer voor het vak bestaat uit 75% examen en 25% maakactiviteiten.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: di-io@tudelft.nl	

Year	2022/2023
Organization	Industrial Design Engineering
Education	Bachelor Industrial Design Engineering

2e jaar Bachelor IO

IOB3-1-22	DP3: Designing System Interventions	10
Course Coordinator	Dr. N.A. Romero Herrera	
Contact Hours / Week x/x/x/x		
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	Dutch	
Course Contents	In dit project maken studenten kennis met de complexiteit van het ontwerpen van producten en diensten (interventies) in complexe netwerken (systemen) van organisaties, producten, diensten, omgevingen en gebruikers. Ze worden er van bewust gemaakt hoe de introductie van, of een verandering in een product implicaties kan hebben op maatschappelijk, organisatorisch en individueel menselijk niveau. Ook leren studenten hoe je daar als ontwerper grip op kunt krijgen. Het project heeft als rode draad het toewijzen van de onderwijsruimtes verspreid over de verschillende faculteiten van de TU Delft campus. Als gevolg van de toenemende, dynamische vraag naar onderwijsruimtes bij de verschillende faculteiten van de TU Delft, staat de TU voor de complexe uitdaging de over de faculteiten verspreide onderwijsruimtes op een efficiënt manier toe te wijzen. Er zal gezocht moeten worden naar een systemische oplossing voor het toewijzen van onderwijsruimtes, passend bij onderwijsvorm en verwachte aanwezig, en rekening houdend met zaken als: de tijd die nodig is om van faculteit A naar faculteit B te komen, financiën en korte-termijnbeschikbaarheid. De universiteit zou graag het thema mobiliteit als uitgangspunt zien en wil daarin zoeken naar duurzame oplossingen voor het zich van A naar B verplaatsen over de campus. Er wordt begonnen met een analyse en een eerste prototype, op basis waarvan de oplossing verder ontwikkeld wordt. Maar ook zullen andere richtingen verkend moeten worden. Denk daarbij bijvoorbeeld aan een digitaal platform voor het reserveren van zalen, real-time toewijzen van zalen, verplaatsen van faculteiten, enzovoorts. Als system intervention designer zul je je eerst vertrouwd maken met het systeem van toewijzen van onderwijsruimtes aan de TU Delft. Dat doe je door individueel in een systeemanalyse (4 weken) onder andere te bepalen wie de betrokken actoren zijn (studenten, docenten, universiteit, facility management) en hun ervaringen met het gebruik van dit systeem in kaart te brengen. Je focus zal liggen op een duurzaamheidsanalyse van de mobiliteit, terwijl je jezelf ook open stelt voor andere ontwerprichtingen. In teamverband zul je vervolgens in drie iteraties (12 weken) gaan ontwerpen. Die iteraties zijn er op gericht interventies uitgaande van mobiliteit te verkennen en dan te beslissen of je daarmee doorgaat of dat je beter andere richtingen kunt onderzoeken. Daarbij zul je van je oplossingen ook de impact op duurzaamheid laten zien vanuit natuur en milieu, sociaal en economisch perspectief. Tenslotte zul je in de laatste 4 weken reflecteren op een interventie die door een ander team ontwikkeld is. Je zult die interventie zorgvuldig analyseren, suggesties doen voor verbeteringen en er ideeën uit halen die je kunt toepassen in je eigen interventies. In de weken 4 (einde van de eerste, individuele opdracht), 8 en 12 (einde van de eerste twee ontwerpiteraties) zullen er ook IDME-activiteiten zijn. Die activiteiten zullen er op gericht zijn om vanuit je design identiteit en je individuele en teamdoelen te reflecteren op de genomen ontwerpbeslissingen en de verkende ontwerprichtingen.	
Study Goals	Als studenten het vak afgerond hebben, zijn ze in staat 1. om geschikte procesmodellen, methoden en technieken te selecteren en deze keuze te onderbouwen voor het uitvoeren van een iteratief ontwerpproces. (LO 1.1.1.) 2. om op verschillende abstractieniveaus te werken voor het creëren van een ontwerp dat een duurzame impact heeft op het milieu, de maatschappij en de economie. (LO 1.2.1.) 3. om zich een mogelijke toekomstige context met betrekking tot een bepaalde situatie voor te stellen, met als doel een gewenste toestand te formuleren. (LO 1.3.2) 4. om concepten te genereren die passen bij een gegeven ontwerpdoel. (LO 1.4.2) 5. om visualisaties, modellen en prototypes te maken voor het effectief testen en evalueren van hun concepten tijdens het ontwerpproces. (LO 1.5.3) 6. om uitspraken te doen over de verwachte duurzame impact van een ontwerpresultaat op het milieu, de maatschappij en de economie. (LO 1.6.4) 7. om een systeemperspectief te ontwikkelen voor verschillende ontwerpuitdagingen. (LO 3.2.1) 8. om complexe probleemsituaties te doorgronden door het onderzoeken van systemische patronen, relaties tussen stakeholders en de variabelen van het systeem waarin het complexe probleem zich voordoet. (LO 3.2.3) 9. om maatschappelijke vraagstukken in het ontwerpen te identificeren en de professionele verantwoordelijkheden van ontwerpers te erkennen. (LO 3.4.1) 10. om relevante argumenten voor hun ontwerpbeslissingen te ontwikkelen en overtuigend te presenteren. (LO 5.2.2) 11. om effectieve communicatiemiddelen te genereren voor verschillende doeleinden. (LO 6.3.1)	
Education Method	Het vak bestaat uit: 1. Een ontwerpproject met zowel individuele als teamactiviteiten langs Q1 en Q2. De teamactiviteiten bestaan uit drie 'design sprints'. 2. Wekelijkse project coaching (Woensdagen/Donderdagen) 3. Wekelijkse weekopeningsessies (Maandagenochten) 4. Drie visualisatieworkshops (elk van twee weken) om visuele prototypingsvaardigheden te ontwikkelen op het gebied van: systeemoverzicht, systeeminteracties en systeem inzoomen. 5. Vrie gastcolleges (Maandagen): systeeminterventies in de praktijk, prototypen van systeeminterventies, duurzaamheidsimpact van systeeminterventies, digitale en fysieke prototyping 6. Vragenuurtjes met experts: visualisaties, duurzaamheid, materialen en productie-technieken, data. Toolkit voor designing system interventions met instructies en templates (te downloaden vanaf Brightspace)	
Literature and Study Materials		
Prerequisites	Alleen studenten die zijn geselecteerd voor de bachelor Industrieel Ontwerpen aan de TU Delft, na het afronden van selectieprocedure, kunnen tot IO bachelor vakken worden toegelaten (artikel 5 - OER Bachelor Industrieel Ontwerpen)	
Assessment	Summative beoordeling Week 4: systeemanalyse (individueel, 15%) Week 16: systeemontwerp sprints (team, 50%) Week 20: systeemreflectie (individueel, 35%, inclusief 10% visualisatieportfolio)	

Enrolment / Application	Formatieve feedback: Week 12: presentatie van sprint 2 (feedback t.b.v. de laatste sprint) Week 18: reflectie op de teampresentatie van de systeemontwerp sprints (feedback t.b.v. de individuele reflectie) For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.
	Registration beyond the registration deadlines is not possible.
Special Information	Contact: dp3-io@tudelft.nl

IOB3-2-22	Product Engineering	5
Course Coordinator	Ir. E.L. Doubrovski	
Contact Hours / Week x/x/x/x		
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor program at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	

IOB3-3-22	Data	5
Course Coordinator	Dr.ing. M.C. Rozendaal	
Contact Hours / Week x/x/x/x		
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course is about DATA and specifically about using data as a design material. What does this mean? Well, we will teach you with a basic understanding of what data is, the different types of data available, its particular qualities, ways of working with it, and how you can use it in your design projects. Many products nowadays use and work with data in their functioning. Think about digital services, devices within the internet-of-things but also robots need data to function. Data is further becoming increasingly important as business intelligence and in the governance of cities. Therefore it is not surprising that design approaches emerge that work with data. Examples are data enabled- or data-centric design. Data is thus all around, as an invisible reality that affects us all, and requires design competence to work with data effectively and responsibly.	
Study Goals	Upon finishing this course students are able to: LO1. explain how data can be used in the design process for understanding, inspiration, creation and validation. (3.2.2) LO2. explain how data connects people, things and organizations in human activities that meaningfully binds them. (2.10.1, 2.10.3) LO3. explain how to make data models by describing data as entities, attributes and relationships. (2.4.4) LO4. explain how to develop a data collection strategy by identifying, comparing and selecting appropriate data sources. (2.4.3) LO5. identify and consider both the merits and risks of working with data in the design process. (3.4.1 , 5.1.1) LO6. explain how to use data visualisation effectively for discovery and communication purposes. (2.7.2) LO7. identify and compare complex data analytical techniques and learn to perform a simple data analysis. (4.2.3) LO8. identify and describe interaction mechanisms and qualities unique to data-driven products and services. (2.7.1)	
Education Method	During the course, we will work with weekly modules, each addressing a specific topic when designing with data. A design context is given that is used in the backdrop of the course providing teaching with a sense of realism and design relevance. You work on these modules individually as well as in teams. For the group work, you will work in the design studios and self-education exercises are set-up by means of online modules and tutorials.	
Literature and Study Materials	A reader will be compiled for this course.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	Each week, your module deliverable (which is a group deliverable) is presented and will be assessed by your teacher based on criteria unique to each module. The feedback you receive allows you to improve this specific deliverable. All module deliverables become integrated in one final group report. This report will be assessed summatively by means of a grade and in cases of unequal contribution to the group assignment the grade may be individually adjusted accordingly. The group report will count for 50% of the final result. A written exam will make up the other 50% of the final result.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: data-io@tudelft.nl	

IOB3-4	Envisioning the Future	5
Course Coordinator	Dr. R. Bendor	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Thinking critically and creatively about the future is an essential part of every successful design process, but what are the best ways to do this? In this course students will learn various methods of what is termed futuring: imagining future designs, contexts, and publics. Students will discover the historical origins and cultural importance of futures thinking, encounter different types of design-relevant futuring techniques, and reflect on the usefulness, limitations, and ethics of applying futures thinking in different design contexts. The course will include three interrelated components: 1. A survey of the historical origins and the social and cultural value of futures thinking. 2. Introduction to a variety of relevant futuring techniques. 3. Opportunities to consider and reflect on the application of futuring techniques in a variety of design spaces and situations.	
Study Goals	Upon successfully completing this course, students will be able to: 1. Demonstrate a sensitivity for the impact of culture on design. 2. Envision a relevant future context for an organisation or a product-service system on organisation, human and technology aspects. 3. Analyse the most important developments and trends regarding technology, organisation and people. 4. Formulate the implications of the analysis in previous LO for envisioning futures. 5. Identify the professional responsibilities of designers within the context of envisioning futures. 6. Assess the value of different types of research and their outcomes. 7. Demonstrate the ability to stay updated in the field of design.	
Education Method	The course consists of weekly lectures and seminar discussions in small groups. Lectures will introduce students to a variety of issues and methods used to envision the future in the context of design. Seminars will provide students with an opportunity to reflect on, and discuss the issues in small groups. Students are expected to come prepared for seminar and be ready to actively participate in discussion. The group assignment (in groups of 4) requires students to research, create, animate and annotate a future vision relevant for an organization (assignment 1) and a product-service system (assignment 2).	
Literature and Study Materials	Readings will be provided through Brightspace.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of the course are assessed through 2 mostly identical group assignments, the first is formative while the second is summative and worth 70% of the final grade. Preparation and active participation in seminars are worth the remaining 30% of the final grade. Active participation in seminar discussions includes preparation (reading assigned material and providing questions for discussion), thoughtful contributions during seminars, and occasional facilitation of discussion.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: etf-io@tudelft.nl	

IOB3-5	Sustainable Impact	5
Course Coordinator	Dr. S.S. van Dam	
Contact Hours / Week x/x/x/x	2-4	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In Sustainable Impact, students learn about sustainability from the perspectives of People, Planet & Prosperity. They develop an understanding of and learn methods for: 1. The complexity and interrelatedness of the UN Sustainable Development Goals (SDGs). 2. How to translate the SDGs into practical priorities for their own design projects using systems thinking. 3. Quantifying environmental impacts using life cycle assessment (LCA). 4. Recovery strategies for products and materials. 5. Sustainable business models and Product Service Systems. 6. Energy effectiveness in products. 7. Influencing user / consumer behavior for sustainability. The content of this course is relevant for students because environmental and social considerations are increasingly crucial for all human endeavors, especially manufacturing, design, and business.	
Study Goals	Upon finishing this course, students will be able to: LO1 - Explain the environmental & social concerns in the UN Sustainable Development Goals (SDGs) (LO 3.3.2, 3.3.3, 3.3.4) LO2 - Identify social issues in design and acknowledge the professional responsibilities of designers. (LO 3.4.1, 3.4.2, 3.4.3) LO3 - Prioritize sustainability concerns & solutions in your own design projects by applying systems thinking methods or tools. (LO 3.3.1, 3.3.2, 3.3.3) LO4 - Calculate environmental impacts using life cycle assessment (LCA). (LO 2.2.2, 3.3.2) LO5 - Assess more sustainable materials & manufacturing methods using LCA, Granta CES Edupack, or other tools. (LO 2.2.2, 3.3.2) LO6 - Identify and ideate recovery strategies for products and materials. (LO 3.3.2, 3.3.3) LO7 - Apply sustainable business models and Product Service Systems to make material recovery strategies economically viable. (LO 2.10.2, 3.3.2, 3.3.3) LO8 - Identify opportunities for energy efficiency, effectiveness, and sustainable energy production. (LO 3.3.2) LO9 - Apply persuasive design tools to drive user / consumer behavior for sustainability. (LO 2.6.3)	
Education Method	The course will start with introduction lectures and exercises on sustainability definitions and systems perspectives, including the UN Sustainable Development Goals, the evolution of Design for Sustainability methods, and the Whole System Mapping method (applied to DP3 and used to frame the following more specific DFS topics). Next, specific DFS topics will be introduced and practiced for six weeks in a row (LCA, material recovery, business models, energy, and consumer behavior). For each of these topics, self-study online lectures and readings will be provided with theoretical elements as well as examples. Q&A sessions with instructors will accompany the self-study modules. Assignments, some interactively workshopped by instructors and some self-study, will guide students to practice the specific tools / methods and apply them to DP3. Workshops will be six 2-hour sessions of 60 students each. Students not participating in DP3 can choose their own product to apply these lessons to, either individually or in self-organized teams.	
Literature and Study Materials	Mandatory study materials: Course Reader Sustainable Impact IOB3-5 will be available as a free PDF downloadable from Brightspace, and/or purchased hardcopy. Design for Sustainability (Open Access): A Multi-level Framework from Products to Socio-technical Systems, 1st Edition Other readings & videos linked to on Brightspace Recommended study materials: Other readings & videos linked to on Brightspace	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	Assessments will be through formative individual quizzes at the end of each two-week session, and one summative individual final exam, covering all course content. During the course, seven formative sets of assignments will be provided which the student can work on either individually or in groups. The assignments will be workshopped during working sessions. These working sessions will have a max of 60 students with two coaches. Formative quizzes will also be supplied to let students self-test on LOs. These quizzes will largely contain questions from past exams, and will be auto-graded in Brightspace. They will be similar to, but not exactly the same as, the final exam. There will be one summative assessment, a final exam, covering all course content through a digital exam on campus.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: si-io@tudelft.nl	

IOB4-1-22	DP4: Designing for Society	10
Course Coordinator	Dr.ir. M. van der Bijl-Brouwer	
Contact Hours / Week x/x/x/x	4 hours/week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course deals with the ability of design to counteract complex global and social challenges in domains such as elderly care, mobility, mental healthcare, safety or biodiversity. There is an increasing demand for such social designers who are by hired by governments, public and social sector organisations and other industries that want to make a positive impact on society. In the course designing for society, you will learn how designers can make an impact in tackling complex societal challenges by meaningfully shaping concepts that address both individual and societal concerns. To achieve this, the course uses the Social Implication Design methodology (Tromp & Hekkert, 2018) as a framework to structure your design approach. The core of this methodology is the development of a vision and design statement that are used to (re-) frame the challenge you are working on, and that guide the development of concepts. In the course we will pay particular attention to the role of insights into experience, meaning and behaviour in a design process. As such, the course combines gathering deep insights into human needs and aspirations, with broad insights into implications for society and the planet. To learn a design approach based on the social implication design methodology, this course divides the approach into four phases, each aligned to one of four assignments: Phase one is focused on an exploration of the domain at hand from an experiential and meaning perspective. Who are relevant actors (people) in the domain? which current experiences are relevant to consider? and what values and meaning underlie these experiences? In an individual assignment, you will learn to create a tangible probe that you can use to explore these phenomena. In phase two, you will form groups in which you will combine the insights from phase one and complement these insights with other context factors to work towards a future context and a design statement which explains how you intend to create change through your design. The future context and design statement collectively make up your design vision. In phase three, you will continue individually to develop and visualize a concept that is in line with your design statement, and to develop an accompanying design rationale. In the final phase, you will develop and build an experiential prototype of one of the concepts developed in your team, to test the assumptions that were part of your design rationale. You will also anticipate and visualize the social implications of your design. In addition to learning a new design approach in this course, you will be introduced to the practice of social design, and you will reflect on the role you want to play as a designer in society.	
Study Goals	LO 1: Students are able to deal with ambiguity and uncertainty during the design process in order to make decisions (1.1.4. Ambiguity and uncertainty) LO2: Students are able to systematically explore the context of a given design challenge from different perspectives to build a clear understanding of the design challenge at hand (1.3.1 Explore design challenge) LO3: Students are able to express critical reflection on their own thinking, decision making and actions and adjust these accordingly, and reflect critically towards the domain (5.1.1 Critical thinking) LO4: Students are able to create visualizations, models and prototypes to effectively reflect in action to develop the design (1.5.1 communicating concepts) LO5: Students are able to balance different interests and concerns to develop a design with optimal sustainable impact that is feasible, desirable and viable (1.2.2 Balance interests) LO6: Students are able to identify social issues in design and acknowledge the professional responsibilities of designers (3.4.1 Moral sensitivity) LO7: Students are able to envision a possible future context related to a given situation to formulate a desired state (1.3.2 Envision) LO8: Students are able to identify and prioritise specific elements of the context to reframe a given proposition and formulate realistic goals (1.3.3 Reframe) LO9: Students are able to work effectively in teams (6.2.1 Work in teams) LO10: Students are able to generate concepts that meet the design goal at hand (1.4.2 Generate concepts) LO11: Students are able to generate effective communication means for various purposes (6.3.1 Communication means) LO12: Students are able to make statements about the anticipated sustainable impact of a design outcome on environment, society and economy (1.6.4 Anticipate impact) LO13: Students are able to position themselves in the field of design (5.3.2 Position yourself)	
Education Method	The educational model to learn to design for society is based on a studio-model which includes coaching, activity-based learning and peer learning. Each student and team will have a coach who will be frequently available to provide feedback on your work so far. To get you started on specific steps in the approach, your coach will facilitate workshop-style activities. Finally, you will also help each other by providing formative feedback in a peer learning structure. To guide your design process, you will furthermore have access to tutorial videos and other resources. In addition, we will offer a number of skill-related interactive sessions where students gain hands-on experience with techniques including storyboarding and team-work. In addition to developing a concept in assignment 3, individual students work on a personal position about the role they would like to take as a designer in society. This personal positioning will be supported by peer support groups and a series of activities in a workshop setting where students will reflect on values, worldviews, and strengths. To collectively explore the role of designers in society we will invite practicing social designers to share their experiences with you.	
Literature and Study Materials	Compulsory reading: Tromp, N., & Hekkert, P. (2018). Designing for society: Products and services for a better world. Bloomsbury Publishing	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	DP4 includes four summative assessments. Assessment 1: individual assignment to create and apply an experiential probe for a selected challenge (4 weeks), with a formative assessment after about 2 weeks. Assessment 2: group assignment on the development of a future context & design statement (vision) for the selected challenge (4 weeks), with a formative assessment after about 2 weeks. Assessment 3: individual assignment on the development and visualisation of a concept based on the developed vision (4 - 5 weeks) with a formative assessment after about 2 weeks. Assessment 4: group assignment on the development and testing of an experiential prototype and an estimation and visualisation of anticipated individual and social implications. In addition, this assignment has an individual component which includes the development of a personal positioning within the design field and within society. Assignment 1: 25% Assignment 2: 25% Assignment 3: 25% Assignment 4: 25% Students who score a course grade between 5,0 and 5,9 (with each separate assignment a 5,0 or higher) can complete a compensation assignment to repair the grade. The maximum grade for the course with a compensated assignment is 6,0.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and	

details.

Registration beyond the registration deadlines is not possible.

Special Information

Contact: dp4-io@tudelft.nl

Year	2022/2023
Organization	Industrial Design Engineering
Education	Bachelor Industrial Design Engineering

Keuzevak Organisations

IOB4-B1-22	Strategic Brand Management	5
Course Coordinator	P. Cankurtaran	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Brands define and differentiate organisations, drive performance and even shape the marketplace. In this elective students will be introduced to the fundamental concepts in brands and branding. They will find out how brand managers cultivate a brand position that resonates with consumers and sets them apart from competition, how brands can inform innovation and how design brings value to the process of branding. They will learn about the strategic and tactical decisions brand managers face in building and managing brands to maintain relevance and fuel growth.	
Study Goals	Upon finishing this course students are able to: LO1. Determine and motivate the positioning of a new brand in relation to opportunities in the market (extending LO2.9.1). LO2. Select, motivate, and develop brand touchpoints to realise the desired brand positioning for a new brand (extending LO2.10.3). LO3. Appraise possible management and innovation options for the brand to propose, motivate and elaborate a meaningful way to fuel brand relevance and growth (extending LO2.9.4).	
Education Method	The course consist of weekly lectures and tutorials, and a project assignment on which students will work as a group (around 4) for the duration of the course. Lectures introduce the theoretical material. Tutorials are a mix of studio discussions and exercises (to promote deeper understanding of the theoretical material and illustrate real world applications of theoretical content), and group feedback meetings and presentations (see Assessment). The brand building project entails the application of course material in the context of a new brand.	
Literature and Study Materials	Books or book chapters, academic/practitioner articles, online learning sources, reports, case texts. Full list of readings and access information will be available on Brightspace.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of the course are assessed through the brand building project report (group; due end of course). The report grade will be adjusted (up or down) for individual contribution based on peer assessment of group members contribution to the process and end result. Students will receive formative feedback on their brand building assignment at the group presentations and feedback sessions. Midway through the course, groups will be offered the opportunity to submit a draft report, should they wish to receive more targeted feedback.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: sbm-io@tudelft.nl	

IOB4-B2	Business Model Design	5
Course Coordinator	Ir. M. Bos-De Vos	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The Business Model Design elective is a course on business models specifically developed for design students. It provides bachelor students with an understanding of what business models are, how they can be designed and what the challenges and opportunities for business model design are. Starting with business models at the firm level, the course will gradually move to business models at the level of an entire ecosystem. Through working on a business model design challenge from practice, students will acquire new knowledge, methods and skills that allow them to meaningfully engage in business model design activities in diverse situations, including new ventures, creative studios and networks of organizations. The business model design knowledge, methods and skills will be instrumental for the projects that students will execute in following courses (such as design projects and master courses in strategic design (Design Strategy Project, Design Roadmapping, New Product Economics)) and for determining the role they wish to have as a (strategic) design professional.	
Study Goals	Upon finishing this course students are able to: 1. Recognize and explain BM definitions and components. 2. Distinguish between different business model archetypes (deepening LO2.10.2) 3. Construct clear arguments for choosing a particular business model design and business model design method. 4. Identify relevant components for a new business model design, and map the value flows to show what the value for each stakeholder is (deepening LO 2.10.1). 5. Construct a business model design for a firm, an interorganizational project, and ecosystem using existing methods. 6. Reflect on critical aspects of business model designs and formulate a statement that addresses or solves a business model design challenge in a particular context.	
Education Method	The course consists of a blended learning approach in which online materials, literature discussion groups, and business model design workshops are combined. The online materials (videos, literature, instructions) are provided per week and can be studied in a self-paced manner. The literature discussion groups (three in total) are held every two weeks. Within this contact time, students discuss and work with the literature via a discussion format that is provided. There is one hour reflection with an expert in the form of a plenary theory discussion. The literature discussion groups allow students to prepare for the exam. The business model design workshops (three in total) are held in the other weeks and focus on a case from practice. During each of these workshops, students will work on the successive parts of their assignment. The instructions for the workshop are provided online and an expert will be available to help with the workshops. Afterwards, students will reflect on the results of their workshop.	
Literature and Study Materials	Mandatory literature & study materials: - Academic book chapters and articles to choose from (per topic). - Videos and complementary materials on business model design methodology. Recommended literature & study materials: - The other book chapters and articles that students did not choose to study. For the literature discussions, students study one of the provided documents, which is mandatory. Through structured discussions with peers, students will also learn about the other documents. Guidelines for the discussions and workshops are provided. The exam questions are focused on specific business model design topics and can be answered by having read different sources.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of the course are assessed through an individual exam and an assignment. The individual exam will consist of multiple choice and open questions and will be 50% of the grade. The assignment consists of four parts: the three business model designs that were developed during the workshops including argumentation (these parts will be delivered with a group), and a business model design statement. The fourth part of the assignment can be developed individually or in a group with maximum 3 students. The full assignment accounts for 50% of the grade.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: bmd-io@tudelft.nl	

IOB4-B3-22	Codesign in Services	5
Course Coordinator	F. Del Caro Secomandi	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This elective course provides an understanding of the service design process with an emphasis on codesign. Different types of organisations, whether in private, public or third sectors, are looking to codesign approaches as a way to design better services that can meet human needs, while distributing benefits and burdens across society in equitable ways. Codesigning services calls for a different way of doing things. Instead of designing 'for' people, its about designing 'with' people. Also, designers must acknowledge that peoples experiences of services can be cocreated by multiple stakeholders interacting through various types of service interfaces, such as rented products, environments and interpersonal encounters. In this course you will study the fundamental concepts and theories about service codesign, learn to identify opportunities to use service codesign in tackling organisational challenges, and apply codesign methods for the creation of new services experiences. You will put knowledge into practice, working in teams to conduct codesign sessions with invited groups of participants. And you understand the empowering role that professional designers can have in service codesign and learn to be more democratic in the making of design decisions.	
Study Goals	Upon finishing this course students should be able to: 1. Explain service codesign concepts and theories in a professional way. (LO 3.1) 2. Identify opportunities to address real-world challenges through service codesign. (LO 1.3) 3. Apply codesign methods to empower people in the creation of new services. (LO 6.1)	
Education Method	The course is divided into two parts. In the first part, students will learn through lectures, self-study, and collaborative work about (1) the main service codesign concepts and theories that are relevant for industrial designers, (2) the main traditions, methods, and tools of participatory design within the field of service design, and (3) the various types of interfaces that can embody services for people, such as rented products, environments, and interpersonal relations. Students will also have weekly lectures by external guests who can provide expert insight into cases, issues, and circumstances of codesign projects in private, public and third sector service organisations. In the second part of the course, students will plan and execute codesign sessions to tackle real-world challenges of service organisations. This will provide students with an opportunity to apply theoretical knowledge in an application domain, for example, health care, mobility, sustainability, etc. For this practical assignment, students will assume full responsibility for organising codesign sessions, eventually with the assistance of teachers and/or external stakeholders to find suitable participants. Throughout part 2 of the course, students will also have regular meetings with dedicated coaches.	
Literature and Study Materials	Stickdorn et al. (2018). This Is Service Design Doing. Sebastopol, CA: O'Reilly Media, Inc. Additional readings will be announced on Brightspace.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The three learning objectives of this course will be assessed in two assignments, as follows: In assignment 1, students will be divided in teams, and each team will (1) do research into a given domain (e.g. health care, mobility, sustainability, etc.), (2) identify opportunities to apply service codesign methods, and (3) plan and execute pilot codesign sessions with invited participants to create new services. The outcome of this work will be presented in a wikipage document that is collaboratively written by team members. Teams will receive feedback from their coaches on first drafts of their wikipages. This assessment is formative and does not count towards the final grade for the course. Then, after delivery of this wikipage, coaches will establish a grade for assignment 1. This grade is summative and represents 30% of a student's final grade. Before starting assignment 2, each student will also perform a self-evaluation on his or her own learning journey, while reflecting on peer-feedback from other team members. This self-evaluation is part of a personal reflection to be delivered at the end of assignment 2, as seen further below. In assignment 2, the teams assembled for the first assignment will reflect on results of their pilot codesign sessions, in order to plan and execute the final sessions with invited participants. The outcome of this work will again be delivered in a wikipage format. The grades given by coaches for these second wikipages are summative and will count towards students' final grades for the course. In addition to the wikipage, in assignment 2 each student will also deliver a personal reflection on their own learning journeys, which builds on the self-evaluation done right after assignment 1. Based on these personal reflections, coaches can add to or subtract from the wikipage grade a maximum of 1,0 point, arriving at an individual grade for each student for assignment 2. This grade will represent 70% of students' final grades for the course.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: cis-io@tudelft.nl	

Year	2022/2023
Organization	Industrial Design Engineering
Education	Bachelor Industrial Design Engineering

Keuzevak Skills

IOB4-S1-22	Design Communication and Visualisation	5
Course Coordinator	Ir. J.W. Hoftijzer	
Contact Hours / Week x/x/x/x		
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The course offers an in depth program of learning profound communication and visualization skills, built on two parallel tiers: (I) generative (including theoretic background) communication/ visualisation skills, and (II) explanatory communication/ visualisation skills. Each student is allowed to adapt his or her program by choosing a specific type of output format as part of tier II: a chosen explanatory communication medium and related skill to learn A fundament of visual literacy (included in this course) will support students in visualizing/ communicating elements of a design situation or process. This visual literacy is needed to be able to unravel and manage information and perception thereof, to visualize concepts, to capture and perhaps simplify a complex situation by communicating it visually. Both tier I and II concern theoretic fundaments and realistic design situations.	
Study Goals	Upon finishing this course students are able to: 1. see, understand (and be able to explain) and accurately apply principles of visual composition (2d and 3d position, orientation, character) to the visualized object/subject (deepening LO 1.1.2, 1.4.3, 1.5.1). 2. effectively consider, justify and apply principles of colour, saturation, brightness, chiaroscuro, when visualizing objects or scenes. 3. accurately and effectively apply generative techniques and methodologies to externalize, reflect and communicate thoughts and concepts (deepening LO 1.3, 1.4, 1.5). 4. apply vigour and signature in their lines, strokes, techniques, to design concepting (deepening LO 1.4.3, 6.3). 5. manually visualize a story: journey, process, narrative, storyboard, using story elements (frames, connections direction, emphasis), focussing on clarity, communicativeness, accuracy (deepening LO 1.3.2, 1.4.2). 6. suitably and effectively utilize/apply specific explanatory visualization techniques: output media (deepening LO 6.3).	
Education Method	The course will be partly individual (tier I), partly group work (tier II). The explanatory communication exercises (tier II) require group efforts: the group aspect will support students in motivating and practically assisting each other, in reflecting on each others work (peer review), and to help each other in the learning effort. Students choose the track they prefer for tier II. Each week, students receive path-specific feedback on their development and achievements. The student has two weekly moments of contact time: (1) for Tier I (in the drawing studios: practice, feedback, active and blended learning), and (2) for tier B (location and method depending on the chosen track). Furthermore, students will spend time autonomously for both tier assignments, and for preparations. It is important to mention that tier I contact time also functions as theory and (indirect) preparation for tier II activities: knowledge about composition, colour, depth, interaction, etc., all benefit both tiers. This structure and approach is in line with the tree of visualization model, in which a strong fundament paves the way for advanced and varying media and applications.	
Computer Use	Laptop and tablet mandatory (in relation to the tiers and chosen tracks) Specific tablet requirements: minimum functional area size 216 x 135 mm	
Literature and Study Materials	Various drawing tools: markers, fineliners, marker paper, t.b.a. Mandatory: Laptop and Adobe Photoshop, tablet see 'Computer use' Recommended literature: Dabner, Grafisch Ontwerpen; Robertson, How to draw, How to render; Eissen, Sketching; Basics and Reader through delftdesigndrawing.com, multiple websites and references to practice. Specific materials for output media tracks to be decided together with external experts.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of the course are assessed through: Formative: In order to know whether improvement or adjustment is needed of teaching, materials and support, the course will contain weekly discussions and moments of feedback between class and teacher. This formative assessment will be done based on students in-between works. Summative: In close relation with the learning goals that were mentioned, the summative assessment will test whether students have attained the required knowledge and skills. Portfolios of both Tiers I and II serve as a basis for the summative assessment. The final grade will be composed based on 65% Tier I and 35% Tier II.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: j.w.hoftijzer@tudelft.nl (dvc-io@tudelft.nl)	

IOB4-S2-22	Leadership and Project Organisation	5
Course Coordinator	A.M. Willemen	
Contact Hours / Week x/x/x/x		
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Leadership and Project Organisation is a course meant for design students. In this elective, students will develop skills required to organise and lead design projects. Students will learn about the aspects of different organisational settings, about leadership styles applicable in a design context, about teamwork and about communication levels. Then, in a series of weekly cases, students will make a project plan and will setup and maintain collaborations within the team and with multiple stakeholders. The relevance of this course for students is that they acquire skills that empower them to realise their goals as a lead designer or project leader in various organisational settings.	
Study Goals	Upon finishing this course students are able to: LO1. to distinguish between various leadership styles and organisational settings and to select leadership styles that fit design and innovation environments (deepening LO 6.2.2.). LO2. to distinguish between communication levels and to select and apply levels according to the situation (deepening LO 6.3.1). LO3. to select and apply techniques to effectively plan and organise design projects and to manage progress of design projects (deepening LO 1.1.3). LO4. to demonstrate collaborating effectively with teams and stakeholders in design and innovation processes in different organisational and cultural settings (deepening LO 6.1).	
Education Method	The course has a blended learning approach, it combines introduction workshops, online material, literature discussions, homework, practice workshops and self-reporting. The online material consists of literature and links to existing videos. The first weeks of the course contain introduction workshops and literature discussions, based on self-study. The 5 sequential practice weeks each start with individual case preparation (based on case description and online video of interview with practitioner) and a brief classroom discussion on Tuesday. On Thursday the practice workshop takes place, guided by a studio coach with input from practitioners. After the workshop, students will fill their progress report and will use that to assess their level of learning.	
Literature and Study Materials	TBD	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	Summative assessment The learning objectives of the course are assessed through an individual report. Students will keep track of their learning in their personal progress report. This report accounts for 100% of the grade. Assessment is based on the quality of their report: does their self-assessment prove by examples that they have applied the theory and learned from the practice, or does it lack level. Formative assessment In weeks 1-8 students will be assessed formatively by means of mentimeter/kahoot quizzes or the like.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.	
Special Information	Registration beyond the registration deadlines is not possible. Contact: lpo-io@tudelft.nl	

IOB4-S3	Mastering Research Methods	5
Course Coordinator	E.A. van den Hende	
Contact Hours / Week x/x/x/x	2,5	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	In this course the student will collaborate with experienced researchers of our faculty on mastering scientific research in design and beyond. The student will discuss published research projects with them, and walk in their shoes during an academic research project Mastering research methods will help the student to develop scientific inquiry skills that are crucial for a plethora of if not all - courses at a university. In any (design) project, it is useful to be able to assess existing knowledge, and decide on the need for gathering new knowledge through research. Moreover, input for solid argumentation typically comes from existing or new research. Knowledge of strengths and shortcomings of research methods is key to stand ones ground. The course offers hands on research experience beyond creating merely awareness. With the help of experienced researchers, this course provides armor against scientifically proven claims, triggers curiosity into the unknown, and shows the merits of scientific research.	
Study Goals	Upon finishing this course students are able to: 1. evaluate research in terms of theoretical and practical implications, limitations, and generalizability (deepening LO 4.1.1, LO 4.3.1) 2. set-up research to conduct an empirical study for answering a research question that matches the topic and practical considerations laid out by the supervisor, and analyze the research accordingly (deepening LO 4.2.1, LO4.2.2, LO4.2.3) 3. present and report research and its outcomes adequately (deepening LO 4.3.2)	
Education Method	The course consists of a kick-off lecture, research expert discussions and an academic research project. The kick-off lecture provides a walk-through of the course and introduces research method basics. The course offers discussion meetings with rotating research experts about published research that employed different research methods. Preparation for these discussions includes watching short videos and studying of background materials about research methods. In parallel runs an academic research project. Research experts formulate the collaborative studio projects and set the conditions. Students can indicate their preferences for a specific studio project (upon availability) and formulate their focus within this project. This is an assignment in duos. For reasons of efficiency and peer support, data collection is likely to be a collaborative effort: multiple duos in a studio contributes to a project database that will be available to all. The research expert supervises a specific studio from the start until the final deliverable. The research expert facilitates regular meetings to discuss project progress, and provides suggestions and support regarding project planning and data analysis.	
Literature and Study Materials	Specific academic papers, depending on the participating research experts in the course. We will detail a reading list one week prior to the course starts and ensure that all can be downloaded from the library and/or are accessible on Brightspace. Mandatory literature: Kumar, R. (2019). Research methodology: A step-by-step guide for beginners. Sage. (fourth edition is also suitable)	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industriel Ontwerpen).	
Assessment	The course includes summative assessment of a short scientific paper (written by duos) about the scientific research project. This determines 100% of the final grade. Formative (peer) feedback prepares for this assessment in three ways: 1) on the prepared discussions of the first weeks, 2) on a short paper writing exercise, and 3) on the draft of the short scientific paper.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: mrm-io@tudelft.nl	

Year	2022/2023
Organization	Industrial Design Engineering
Education	Bachelor Industrial Design Engineering

Keuzevak Technology

IOB4-T1-22	Materials and Manufacturing	5
Course Coordinator	S. Ghodrat	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this elective, students will experience that materials are not only items in a data base that need to be selected, but that they can also mix and combine e.g. bio-based and degradable components to create their own materials. Why not using a starch-based material to make an edible lunch box, or designing your own bio-based cover for your tablet; why not using maizena-sand mixture to make a flower pot or a mussel shell composite to 3D-print a ceramic lamp? The use of bio-based materials is at the core of this elective. Students will learn appropriate tools which are needed to design with such new (and often not yet existing) material composites. How can you best model, make and test a product made by a new material designed by yourself? Manufacturing, testing and user aspects of the case studies will be topics in this elective. The content of this course is relevant for students, because they will gain knowledge and insight on how selection of appropriate manufacturing techniques and of materials will condition the design of a product. The performance (technical and experiential) of finished products is ultimately determined by underlying materials and manufacturing characteristics.	
Study Goals	Upon finishing this course students are able to: 1. Categorise the materials based on understanding their common properties and characteristics as well as variety of manufacturing technologies and the related opportunities and limitations for embodiment design (deepening LO2.2.2). 2. Measure and evaluate the basic mechanical properties of materials with e.g. tensile test, bending test and hardness. In addition, they will become familiar with experiential characterization of their materials used in their case studies (deepening LO 2.2.2). 3. Describe and analyse both design possibilities and technological limitations for their selected manufacturing technologies in their embodiment design, including the peculiarities of bio-based materials (deepening LO 2.2.2). 4. Use and analyse modelling (SolidWorks+ Finite Element Analysis) of more complex shaped products and calculate the loaded product and production aspects such as feasibility (deepening LO 2.1.2). 5. Recommend a suitable material and manufacturing method in a given design situation, taking into consideration a given set of boundary conditions and sustainability aspects (deepening LO 3.3.2).	
Education Method	The course consists of lectures, workshops and a project assignment. In a series of lectures, students will be taught about materials in general (wk1+2), SolidWorks with Finite Element simulations Add-in (wk 3), manufacturing (wk 4+5), technical and experiential characterization (wk 6+7). Each lecture is followed by a workshop on the lecturing topic, including a measurement practicum (wk 6) and a lab visit with demonstrations of advanced testing (wk 7). During the course groups of students work on a case study e.g. a lunch box, which will involve aspects of material selection, characterization, shape redesign and finite element simulation. Workshops: During the weekly 2-hour workshops, students in groups of 3 will work on their case studies as project assignment. This provides students the opportunity to practice the theory, while they document their process and findings in the final report and present it in a video clip.	
Literature and Study Materials	Mandatory Literature: The course reader containing different chapters based on topics covered in this course. Different chapters from different books will be used as sources: For materials and testing: e.g. M. F. Ashby, Materials, engineering, science processing and design, Elsevier Ltd, 4th edition 2019. Chapter 2 (Family trees: organizing materials and processes), Chapter 5, Chapter 6, Chapter 9 (cyclic Loading, Fatigue) W.D. Callister, Jr., Materials Science and Engineering, 9th Edition, John Wiley & Sons. Chapter 1 (Introduction: Classification of Materials), Chapter 8 (Mechanical Properties of Metals), Chapter 10 (Fracture Toughness Testing and Fatigue), Chapter 13 (Properties and Applications of Metals), Chapter 14 (Brittle Materials and Flexural strength), Chapter 15 (Characterization and Applications of Polymers). For manufacturing: A set of videos and some lecture notes with references and CES Edupack. Software CES Edupack (for both materials and manufacturing) SolidWorks Simulation, (FEM Add-in) Recommended literature: Internet websites e.g. Wikipedia, video tutorials, related references and papers.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of the course are assessed through a group report and an exam. 1. Group work (60%): mini report (4 page- report with a defined format) + video clip (3-5 min). 2. Individual (written) exam (40%): Multiple choice questions (on computer) and descriptive design question(s) (on paper). Formative assessment: There will be a quiz at the beginning of each workshop from the topic of the previous lectures. In week 4, there will be a group presentation as a progress report.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: mm-io@tudelft.nl	

IOB4-T2	Mechatronics	5
Course Coordinator	Dr. Y. Song	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Mechatronics course offers unique opportunities for students to design smart, affordable, reliable and versatile consumer products. For this, students will learn knowledge and skills related to electronics, digital signal processing, control, (data-driven) modelling, simulation, prototyping, as well as mechatronic design principles, where the key is to achieve synergy between mechanical and electronic components while striving for added value, e.g. economical and user benefit, in consumer products. Based on prior knowledge on dynamics, electronics and software, and envisioning the future of smart product design, the mechatronic course will further explore the design of mechatronics products by focusing on three key questions: 1. What are the working principles of mechatronic system and how can (micro) electronics be applied in the system? 2. How can Mechatronic design principles be applied in designing mechatronic products and how do these principles affect the product? 3. How can we implement and optimize the performance of mechatronic system by combining the knowledge on physical models and electronics using intelligently balanced control strategies?	
Study Goals	Upon finishing this course, students are able to: LO1: Explain the principles of (micro) electronics for sensors, sensor circuit, digital signal processing, motors, motor control circuit and embedded system (deepening LO 2.3.1); LO2: Design a simple (PID) controlled mechatronic system and select proper components/algorithms for specific applications; LO3: Model and simulate such as a mechatronic system and explain how different parameters influence the behaviour of the system (deepening LO 2.1.2, 2.3.2); LO4: Implement and optimise the system using a (PID) controller to meet the desired needs within realistic boundary conditions; LO5: Assess the implication of mechatronic design principles on the performance of products (deepening LO 2.3.2);	
Education Method	The course consists of lectures, workshops and a project. The Lectures with practical examples (e.g. case studies by external specialists) are held once per week; The workshops (in groups) are held once per week in the first 6 weeks on the topics of the lectures, respectively A project assignment with the focus on mechatronics system.	
Literature and Study Materials	Lecture slides, specified internet resources, The Mechatronics Handbook (Free) will be used as a reference	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of the course are assessed through a group report and an exam. The group report has a maximum of 10 pages (with a video) and will be 50% of the grade. The individual exam will consist of multiple choice and open questions and will be 50% of the grade.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: mechatronics-io@tudelft.nl	

IOB4-T3	Machine Learning for Design	5
Course Coordinator	A. Bozzon	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This technology elective will provide students with the knowledge required to understand, design, and evaluate machine learning systems in the context of the design of intelligent products, services, and systems (iPSSs). Machine learning (ML) is a computational approach that aims at giving computers the ability to learn without being explicitly programmed (A. Samuel, 1959). Smart thermostats, voice-based personal assistants, autonomous vehicles, traffic control systems, online social networks, web shopping platforms, content creation platforms, personal health appliances: much of current and future iPSSs are powered by ML technology, influencing and shaping our interests, habits, lives, and society. To meaningfully envision and design future iPSSs that is beneficial and useful to people and society, designers must: engage with the details of how ML systems see the world, reason about it, and interact with it; experience the quirks, biases, and failures of ML technology; contend with how agency, initiative, trust, and explainability mediate the interaction between human and iPSSs; and understand how functionalities enabled by ML can be designed in iPSSs. Students in this course will gain practical experience with ML technology and learn how to think critically of what ML systems can do, and how they could and should be integrated into iPSSs. The course will cover the following topics: 1. AI and ML in intelligent products, services, and systems (design). The students will be introduced to the role that Artificial Intelligence and, more specifically, Machine Learning technology play in iPSSs, in their design process, and in their lifecycle. The most important ML concepts, terminology, paradigms, and methods will also be described and exemplified. 2. Text Processing methods for iPSSs. Building on the knowledge acquired in the DATA course, students will be introduced to existing ML techniques for text processing (parsing, analysis, generation), and will practice with their use in realistic design contexts. 3. Image Processing methods for iPSSs. Students will be introduced to existing ML techniques for image processing (acquisition, analysis, generation), and will practice with their use in realistic design contexts. 4. Creating Machine Learning Models Students. will deepen their knowledge in how machine learning models could be trained, and on how their performance could be evaluated. Specific attention will be devoted to the task of training data a) design, b) creation, and c) usage in realistic adoption scenarios. 5. Designing iPSSs that include Machine Learning technology. In this final module, students will learn how to design (using human-centred approaches) intelligent products, services, and systems. Students will understand which issues could impact the experience of people when interacting with iPSSs, and learn to assess how ML technology can and should be integrated into iPSSs.	
Study Goals	Upon finishing this course students are able to: - LO1: explain common terminology used when discussing about Machine Learning algorithms, systems and their evaluation (deepening LO 2.4.4) - LO2: identify the requirements for a Machine Learning system (deepening LO 2.5.2) - LO3: describe and compare different classes of Machine Learning techniques (deepening LO2.4.1, LO 2.4.4, LO 3.2.3) - LO4: describe and compare design decisions in the context of iPSSs design and defend the relevance and correctness of their choices (deepening LO2.5.1, LO 5.1.1) - LO5: recommend appropriate Machine Learning technique(s) for a certain problem domain (deepening LO1.2.2, LO1.2.3, LO3.4.3, LO 5.2.2) - LO6: assess their team members' work with respect to methodological soundness and contribution. (deepening LO1.5.4, LO 1.6.1, LO 5.1.2)	
Education Method	After a first introductory week, the course will include four modules, each including: (1) two lectures (one per week) where the main theoretical concepts related to the module are introduced and put in context; (2) studio work: during the first week of a module, students will have individual hands-on experience with the technology related to the module, while during the second week students will collaborate in teams to work on their bi-weekly formative assignments (teachers and student assistants are available for supervision and feedback), and (3) self-study based on readings and additional teaching and practical material associated to each module, and related individual formative assignments.	
Literature and Study Materials	Students will be provided with a reader, containing curated readings, lecture notes, tutorials, and Webcasts related to each course module. For individual hands-on experience, students will be provided with examples (templates) of practical assignments. These templates serve as references for the group assignments constituting the assessment elements of the course. Practical work will be performed through interactive web-based resources and online visual programming tools selected to facilitate understanding of ML techniques and models.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of the course are assessed through an individually assessed group work, and an individual exam. Group work (50%): students will be organized in groups of 4 or 5 members. Groups deliver one report for each course module to build a final portfolio of group work that is assessed at the end of the course, and that accounts for 80% of the group work. Students will receive an individual grade for the group work. The individual grade includes an individual assessment - that takes into account input from group members - which accounts for 20% of the group work. Individual exam (50%): a mix of multiple choice and open answer questions. Groups will attend weekly formative coaching sessions. Weekly Individual formative assessment will allow students to test their knowledge of the topics addressed in a bi-weekly module. Multiple choice questions will be automatically evaluated, open answer questions will be evaluated by teaching assistants.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: mlfd-io@tudelft.nl	

Year	2022/2023
Organization	Industrial Design Engineering
Education	Bachelor Industrial Design Engineering

Keuzevak People

IOB4-P1	Human Factors & Ergonomics	5
Course Coordinator	Dr. T. Huysmans	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this elective course the students knowledge of human factors and ergonomics (HFE) is deepened. The HFE discipline takes a human-centered and scientific approach to the design of products, services, and systems in a way that balances system performance and human well-being. This course will guide students in acquiring knowledge on the complexities of the human system in terms of physical, perceptual, and cognitive capacities. These capacities will provide the fundamental boundaries that form the design space in which one can be creative. The course will provide the theoretical backgrounds in the human systems, how the capacity of this system can be measured, and how these measurements can be used in a systems approach to the design of product-service systems. Case studies will be in the field of healthcare, mobility, sports and consumer product design. The following topics are covered: 1. Physical, perceptual, cognitive, and organizational factors 2. Measurement and research methods 3. Systems approach and HFE design 4. Digital human modeling and simulation 5. Various applications This course is relevant for the student as it provides a deeper insight in the complexities of humans and the challenges of considering human factors during the (re)design of products/services/systems in key application domains.	
Study Goals	Upon finishing this course students are able to: 1. Explain the value of the HFE discipline for design of products, services, and systems. (deepening LO2.6.1) 2. Explain the human physical, perceptual, cognitive, and organizational complexities in the context of socio-technical systems. (deepening LO2.6.1) 3. Explain and apply HFE research methods to analyze HFE challenges at different levels (component, system, organization). (deepening LO 1.1.2, 4.2.2) 4. Apply a systems approach in the design process to balance safety, performance, and satisfaction. (deepening LO3.2.1) 5. Apply computer models in HFE simulation 6. Critically analyze the context, present, and justify, with scientific rigor, the application of HFE solutions in mobility, healthcare, sport or consumer product design deepening LO 5.1.1, 6.3.2.	
Education Method	The course consists of two main educational parts: lectures with corresponding workshops and a group project (5 students). Lectures and workshop: The lectures are mainly focused on setting the stage of the human factors discipline: What is HFE? How to do a HFE study? What are the complexities of the human system and how to design for them? In what domains is HFE applied? How to communicate about HFE? This is done by providing the students with a reader, a set of lectures, and quizzes to test their understanding. Most of the lectures are during the first weeks of the course. Each of the lectures focusing on a specific factor (physical, perceptual & cognitive, and organizational) is followed by a workshop where taught theoretical concepts are further deepened by the student in a hands on way, e.g. by (computer) exercises, group discussions, and feedback moments. Group project: from the third week, students get hands-on experience with human factors on a specific group project where they have to analyze an existing product/service/system from different HFE perspectives, do an evaluation study, and come up with redesign recommendations. Via the group project the student sets theory into practice, but their project also requires them to further deepen their knowledge independently. This process is steered via weekly coaching.	
Literature and Study Materials	A reader will be compiled as a selection of several chapters from different books and scientific articles. In addition, a collection of online resources will be provided as a starting point for the student to further deepen their knowledge on physical, perceptual, cognitive, and organizational human factors as required for their group assignment. Instructional videos are provided to demonstrate relevant tools and techniques in HFE (HFE Toolbox). Finally, a number of podcasts (released in Week 3) will be shared where we interview experts on the topics of the group projects.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industriële Ontwerpen).	
Assessment	The learning objectives of the course are assessed through a group project (report and presentation) and an individual exam. There are two deliverables for the group assignment (60% of the grade). 1. A scientific report (12-page paper) on HFE-based evaluation of a product/service/system, delivered in Week 9, and 2. a 10-minute presentation making a compelling case for the HFE-based evaluation and redesign recommendation that is targeted at a specific stakeholder and builds on the findings in the paper, delivered in Week 9. The individual exam will consist of multiple choice and open questions and will be 40% of the grade. Formative assessment is achieved through BrightSpace Quizzes before and after lectures and polls during the lectures. Coach, expert, and peer feedback is provided on progress and report throughout the group assignment and on the interim report in week 6.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: hfe-io@tudelft.nl	

IOB4-P2	Experience, Motivation & Behaviour	5
Course Coordinator	Dr. J.J. Kraal	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	In this elective, students will develop a deeper knowledge of human behavior, motivation and experience. A focus will be on understanding how to influence individuals, social and environmental factors as well as tools and strategies to design with. During the course, we will apply the principles of psychology and behavior change on the themes of health, mobility and sustainability. The content of this course is relevant for students, because it will help students to get a better understanding of how design can support people to change their behavior.	
Study Goals	Upon finishing this course students are able to: 1. Explain which factors influence motivation and behavior using frameworks from psychological theories covered in the course (deepening LO2.6.3). 2. Analyze existing behavior change interventions using motivation and behavior frameworks covered in the course. 3. Apply principles of motivation and behavior in their designs to measure, change and evaluate motivation and behavior (deepening LO2.6.3). 4. Apply principles of human experience, emotion and perception in their design. (deepening LO2.6.2) 5. Evaluate design interventions and relate these results to theoretical frameworks that are discussed in the course. 6. Substantiate clearly their insights and design decisions during a presentation and in a written report (deepening LO 6.3.2)	
Education Method	The course consists of two main educational parts: lectures and a group assignments. The lectures will provide students with the information regarding the basic principles of experience, motivation and behavior, and will be performed by different lecturers and guest lecturers. In the assignments, student will apply the content that is provided in the lectures. During the assignments, the students will be supervised by a coach giving direction on their work and deliverables. During the coach meetings, students will present their work and discuss the topic with their peers.	
Literature and Study Materials	A reader will be created with chapters from several books and scientific papers. The book chapters will focus on content, explaining theoretical frameworks and models. Scientific papers will provide examples on how these frameworks are applied in design interventions on different domains and target groups. Recommended literature and study materials: - Kristina Niedderer, Stephen Clune and Geke Ludden (2018) Design for Behaviour change: theories and practices of designing for change. Routledge, NY - Stephen Wendel (2013) Designing for Behavior Change: applying psychology and behavioral economics. O'Reilly Media, Inc.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of the course are assessed through a written report and presentation performed in the last week. The report and presentation provide an overview of the assignment, the insights it created and a critical reflection on their own work. The final grade will be calculated based on the written report (80%) and the presentation (20%).	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: emb-io@tudelft.nl	

IOB4-P3	Culture & Society	5
Course Coordinator	Ir. A.G.C. van Boeijen	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	In this course students will learn how to study cultural and societal processes. The social and cultural perspective will empower students to move beyond the individual as the unit of design. Through close readings of text and design case studies, we will focus on the relationships between people, and between people and their environment. The intention of most contemporary designers is to contribute to changes for better futures. This course critically reflects on this concept of change. We will raise and address the questions about what exactly should be changed and what not, for whom these changes are designed, how change happens over time and place, and what the possible roles of the designer in relation to change might be. Insights from this course will inspire the development of students design visions and help them communicate the outcomes of their socio-cultural analysis. The content of this course is relevant for students, because what we design shapes cultures and societies and vice versa: their logics and values influence our designs.	
Study Goals	Upon finishing this course students are able: 1. to analyse the anthropological, sociological, political, and economic context that stage and shapes society. (2.8.1/2.8.2) 2. to identify social and cultural issues in design. (3.4.1) 3. to critically reflect on cultural and societal issues. (5.1.1; 5.4.2) 4. to communicate their analysis on culture and society. (6.3.1)	
Education Method	The exchange of thoughts and opinions is essential for critical thinking. Therefore, we choose education methods that enhance the interaction between students, and between students and experts of theoretical concepts. The course consists of weekly activities: Lectures for the explanation of theory and time for Q&A; Readings for self-study; Seminars in small groups to reflect and discuss key concepts; An assignment resulting in a representation of insights in a portfolio; Intermediate feedback moments (by peers and lecturers). A blended format (combination of offline and online activities, e.g. video lectures and e-feedback) is offered to support both individual learning and learning through discussions in class.	
Literature and Study Materials	The course will use a reader with papers, book sections, and links to documentaries.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	During the course of the course, partial individual deliverables (as part of the portfolio) are requested, on which formative feedback is given by both peers and lecturers. At the end of the course, the learning objectives of the course are summatively assessed through individual work. The grade will be calculated based on this individual final work.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: cs-io@tudelft.nl	

Year	2022/2023
Organization	Industrial Design Engineering
Education	Bachelor Industrial Design Engineering

3e jaar Bachelor IO

IOB6-1-22	DP5: Bachelor Final Project	15
Course Coordinator	G.S. Celik	
Course Coordinator	Ir. G.P.M. Hoekstra	
Course Coordinator	S.M. Persaud	
Contact Hours / Week x/x/x/x		
Education Period	1 4	
Start Education	1 4	
Exam Period	none	
Course Language	English	
Course Contents	In the Bachelor Final Project (BFP) students apply the knowledge and skills they have gathered and competences they have been working on so far in the bachelor. In this way the project can be seen as a 'Proof of Competence' for the whole Bachelor programme. At the same time the project is part of the learning trajectory to become a BSc Industrial Design Engineer and therefore the directional activities are corresponding with the Final Attainment Levels. BFP (Dutch: Bachelor Eind Project (BEP)) revolves around a design assignment for a real client, with the exploration and application of (new) knowledge in context. This project requires students to learn: - individual cooperation with a client, - to plan and anticipate, - to be sensitive to their own project and performance, and - to be willing and able to correct the project if required. Effectively, this means a student needs to (learn to) individually manage the relation between cause (effort) and effect (results) and act responsible. Participants in BEP will be offered specific workshops as support for their design process and presentation. Once started, each student develops their own design goal and project goals within the given case, using the Navigation Canvas (Van Erp et al 2020). The student also sets up a project structure and planning accordingly, including: - strategies for integrating knowledge, - application of design tools and methods, and - fruitful cooperation. Students can use the Navigation Canvas (Van Erp et al 2020) to encounter any typical phase of the product development process in BEP: - the fuzzy front end (strategy building), - ideation and concepting, embodiment and - the fuzzy back-end (implementation plan). Depending on the given case and the goals of each individual student, focus and emphasis can lay on different phases and elements. The set up for the project and its deliverables are agreed upon by the design coach and the client. Based on the shifting field of work, the methodology of the course will be based on classic methodology via Design Thinking to Systemic Design Thinking. Three times during the project the achieved results and plans are presented to and discussed with the client. At the end each student should deliver a credible design proposal that offers value for all relevant stakeholders.	
Study Goals	Upon finishing this course students are able to: 1. Navigating Students are able to apply appropriate process models, methods and tools to navigate an iterative design process (1.1). 2. Integrating Students are able to integrate the interests of different stakeholders to generate designs that have sustainable impact on environment, society and economy (1.2). 3. Framing Students are able to explore complex problem situations and use the insights to frame (open) design challenges (1.3). 4. Concepting Students are able to explore the design space and generate innovative concepts (1.4). 5. Visualising & Prototyping Students are able to make concepts tangible through visualizing, modelling and prototyping to develop, communicate and evaluate designs (1.5). 6. Evaluating Students are able to evaluate design outcomes with regards to feasibility, desirability and viability as well as the sustainable impact on the environment, society and economy (1.6). 7. Professional Development a. Students are able to explain the global field of design and to position themselves in this field (5.3). b. Students are able to set up a plan to develop themselves towards a preferred learning and working career with a high degree of autonomy (5.4). 8. Collaboration & Communication a. Students are able to initiate and maintain various collaborations effectively to fulfil a specific purpose within the design process (6.1). b. Students are able to present and report research and design outcomes using a variety of media (6.3).	
Education Method	The Bachelor Final Project is a full quarter full-time (15 EC, 420 hours) project with a real client. The project is performed and executed by students individually but they will be supported by a team of two design coaches per case. Each project case will function as context for about 16 individual projects. Leading in the project are four planned meetings with the client: Company visit, Framing, Co-Working, Final Result Presentation. Coaching is planned and executed at least once per week for every student. Scheduling by coach and student is in consultation. Before the kick-off of the project, students enroll for one of the project cases presented on Brightspace, by ranking them all according to preference. The first two weeks will be dedicated to the research with common planning and division of tasks, of which the research results will be common knowledge. The course as a whole starts with an introduction lecture with contributions by former BEP students. The course is concluded by the final assessments and a final evaluation session with the design coaches.	
Literature and Study Materials	TBD	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	

Assessment	Admission requirement apply for the Bachelor Final Project (article 14 - OER Bachelor Industrieel Ontwerpen). Only students can be admitted who, at the moment of admission (note: this is earlier than the start of the course), have fulfilled the requirements. Check the OER for the exact admission requirements. In the Bachelor Final Project students are assessed summative on all Learning Objectives of BEP at the end of the course. The students are assessed formative at midterm on their framing of the content of the project brief and the quality of the first design directions. This midterm assessment is divided in two parts: a peer assessment and an assessment by a committee of coaches and clients. To support the assessors and safeguard a consistent final assessment across all assessors we make use of a rubric with assessment criteria defined at four levels. Coaches also keep a coaching discussions (journal) during the project. The deliverables for each student for BEP: 1) Mid-term report (formative) 2) Final report (summative) 3) Final design (summative) 4) Reflection (summative) The weight of each deliverable in the final grade is determined as: Final Report 60% Final Design 30% Reflection 10% The final grade of each student is provided by a committee of assessors. For this project a final grade of 6.0 or higher and sub-grades of 5.0 or higher, are required for passing the course. The assessment is provided to the students by the use of a course-specific assessment form and through feedback and evaluation meetings between student and coaches. Enrolment / Application For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. Special Information Contact: dp5-io@tudelft.nl
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Year	2022/2023
Organization	Industrial Design Engineering
Education	Bachelor Industrial Design Engineering

Keuzevakken

IOB6-E1	Design for the Circular Economy	5
Course Coordinator	Ir. I.C. de Pauw	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Design will play a key role in the transition to a circular economy (CE). Most of today's products are designed with little regard for what happens when we no longer want them. Consequently, products are wasted, in terms of both materials and value. And waste is design gone wrong! In a circular economy, products loop back into the production system through repair, refurbishment, remanufacturing and eventually, recycling. By doing this, we extend their useful life, capture value and design out waste. This course will strengthen the design skills of bachelor students to tackle the transition from our linear 'take-make-waste' economy to a circular economy. This elective course provides students theory as well as practicum assignments, addressing the business and design aspects of going circular. The student will learn about and develop circular business models and recovery strategies applied to a specific design challenge.	
Study Goals	Upon finishing this course students are able to: 1. Describe the basic principles and terms underlying the circular economy (deepening LO 3.2). 2. Explain business models and recovery strategies that can deliver value in the circular economy (deepening LO 2.10.2 + 3.2). 3. Evaluate the circularity and sustainability of a product (deepening LO 1.6.4). 4. Apply CE principles to redesign a (linear) product, addressing both the business case and recovery strategies (deepening LO 2.10.2, 3.2, 3.3.1, 3.3.2). 5. Provide relevant arguments for your design decisions, using the theory, methods and tools offered (deepening LO 5.2.2).	
Education Method	This course is blended, consisting of the MOOC Circular Economy, lectures and practicums. The MOOC and lectures cover the theory of the course, and the MOOC offers formative assignments for students to test their understanding of the theory. During the practicums, students translate the theoretical knowledge into design practice and work within groups on a design project (resulting in a group report). Feedback is provided during the practicums by content experts and group coaches. The course ends with an individual, written assignment on the knowledge addressed in the MOOC, lectures and practicums.	
Literature and Study Materials	Mandatory Literature & Study Materials: All study materials are offered via the MOOC and Brightspace, including videos, lecture slides and selected papers. Recommended Literature & Study Materials: Bakker, C. A., Den Hollander, M. C., Van Hinte, E., & Zijlstra, Y. (2014). Products that last: Product design for circular business models. BIS Publishers.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The grade for this course is based on two summative assessments: - An individual, written assignment on the knowledge addressed in the MOOC, lectures and practicums (addressing learning objectives 1, 2 and 5). - A group report describing the results of the design project (addressing learning objectives 3, 4 and 5). The final grade is determined by 50% of the grade for the individual assignment and 50% of the grade for the group report.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: dfce-io@tudelft.nl	

IOB6-E10	Design Didactics	5
Course Coordinator	Ir. J.J.M. Zijlstra	
Contact Hours / Week x/x/x/x	4 + stage in DP2	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Course Contents	Bij Ontwerpididactiek kunnen studenten de basisbeginselen leren van didactiek en hun kennis van het ontwerpproces en ontwerpmethoden verdiepen. Ze ontdekken hoe ze hun inzichten het beste aan medestudenten (IO) kunnen overbrengen. Het keuzevak Ontwerpididactiek bevat onder andere de volgende onderdelen: Didactiek: theorie over het coachen van het ontwerpproces van medestudenten; Stage lopen bij DP2, waarbij de student ervaart, voelt, ontdekt, experimenteert met wat er bij de theorie is behandeld; Verdieping van de theorie m.b.t. ontwerpen, ontwerptechnieken en ontwerpmethoden. Door het evalueren van video-opnamen van hun eigen workshop en tijdens begeleidingsgesprekken kunnen studenten hun vaardigheden op deze onderdelen verder aanscherpen. Parallel hieraan komen ze elke week terug in een intervisiebijeenkomst waarin met medestudenten wordt besproken wat ze in de praktijk hebben geleerd. Studenten die dit vak met goed resultaat afronden kunnen in aanmerking komen om studentassistent te worden bij IO-vakken.	
Study Goals	Als studenten het vak afgerond hebben, kunnen ze: 1. Zelfstandig studenten begeleiden door middel van gestructureerde gesprekken; 2. Door het toepassen van de didactische theorieën de ontwerpprocessen en methoden inzichtelijk maken voor andere studenten (verdieping leerdoel 1.1.1); 3. Een groep studenten effectief en motiverend instrueren, onder andere d.m.v. een workshop; 4. Studenten helpen zelf na te denken en verder te helpen bij het oplossen van een voorliggend probleem (verdieping leerdoel 1.1.2); 5. Het werk van de door de student begeleide studenten beoordelen a.d.h.v. een gegeven beoordelingsmodel; 6. Effectieve feedback geven op de didactische aanpak van collega-studenten (verdieping leerdoel 5.1.2); 7. Met persoonlijke diepgang reflecteren op hun eigen gedrag en de verkregen inzichten omzetten naar verbeterstrategieën (verdieping leerdoel 5.1.1).	
Education Method	Het vak bestaat uit: Theorielessen en workshops: hierin wordt de theorie behandeld en geoefend die van belang is voor de coaching van de komende week; Stage in het DP2-project: onder toezicht van een vaste studiodocent passen de studenten de theorie toe door studenten te begeleiden in hun ontwerpproces, maar ook door het geven van minimaal één workshop en door te ondersteunen bij de beoordeling. Wekelijkse intervisie-bijeenkomsten: tijdens deze bijeenkomsten van een uur delen de studenten ervaringen en helpen ze elkaar om oplossingen te bedenken voor lastige of vaak voorkomende situaties.	
Literature and Study Materials	Delft Design Guide. Verder wordt literatuur aangeboden in de course manual.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen). Dit vak mag alleen gevolgd worden door studenten die het eerste studiejaar en DP3 succesvol hebben afgerond of door studenten van cohort 2020 en eerder die minimaal 60EC en 3 van de Design Projects uit het eerste en tweede jaar hebben behaald. Er wordt geselecteerd op basis van motivatie i.v.m. beschikbaarheid plaatsen. De motivatie kan worden gestuurd naar dd-io@tudelft.nl tot en met de eerste zondagavond van januari.	
Assessment	De toetsing is individueel en gebaseerd op het in te leveren eindrapport en twee in te leveren video's. De beoordeling van de DP2-studiodocent wordt ook meegenomen in de eindbeoordeling.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: dd-io@tudelft.nl	

IOB6-E2	Reimagining Mobility	5
Course Coordinator Contact Hours / Week x/x/x/x Education Period Start Education Exam Period Course Language Course Contents	Dr. E.Y. Kim 4 3 3 3 English Reimagining Mobility is an elective design course that teaches the 3rd year IDE undergraduates how to envision meaningful mobility solutions in different levels of innovationproduct, service, or societal-system levelsthat harmonize people and different transportation modalities. The content of this course is relevant for students who are interested in mobility theme, because through an iterative design process, they will be equipped with the domain knowledge on general mobility sector (in product/ service/societal-system levels), an ability to apply the previously acquired design mindset and skill sets (from other design courses in the faculty) to analyze real-world problems in the mobility sector, and an ability to design new mobility concepts for people in transit and stakeholders in the mobility ecosystem. The course will be instrumental for the projects that students will execute in master courses and for determining the role they wish to play as design professionals in mobility related societal challenges.	
Study Goals	Upon finishing this course students are able to: LO1. Analyze transportation modalities of present context (e.g., autonomous driving, busses, trams, trains, aviation, airport to micro-mobility, etc.) LO2. Create a meaningful mobility solution built upon emerging technologies, making use of the design mindset and skill sets acquainted throughout the previous years LO3. Explain the novel ways by arguing how the future mobility solutions newly interact with stakeholders in the mobility ecosystem LO4. Describe design knowledge across the iterative design innovation process and explain their design journey.	
Education Method	The course consists of lectures, in-class team activities, guest lectures, open discussion, and a project assignment. Lectures are held once a week, and provide the student with information on how to execute the project assignment, also some guest speakers will be invited. In-class team activities (in groups of three~four) are held every week, and provide students the opportunity to practice with the real-world problem finding and solving. Project assignments are done in groups, and consists of making a final solution of a digital advertorial (advertisement + editorial). Guest lectures by internal faculty members and external professionals will be held throughout the quarter.	
Literature and Study Materials	Readings will be announced on Brightspace.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of the course are assessed through 1. Mid-term presentation by each student team, and 2. Final showcase/exhibition. 1. Mid-term presentation (by each student team) - formative During the lecture hour, students give a 10 minutes lecture about their project progress to their coach and the other teams in the same coaching group by demonstrating the understanding of the present mobility context in relation to their own definition of its meaningfulness (week 2), building upon the activities completed in Week 3 (personas, empathy maps) and Week 4 (TRL and design requirements). The mid-term presentation will be assessed formatively (no grading, only feedback for improvement). The mid-term presentation addresses Learning Objective 1. 2. Final showcase & digital advertorial 100% of the final grade - summative At the end of the course, student teams will present their final mobility concepts in the format of the advertorial (which consisted of short write ups, any supporting image, framework, prototype, brochure, or video). During the final showcase, the advertorial will be presented to the rest of the teams in the class, mentors/coaches. The final showcase addresses Learning Objective 2, 3, and 4. Note that students in the same group will get the same grade but we will ask each student justify the individual contribution to prevent students to free ride. Detailed scoring rubrics and information on what should be included in the final showcase will be provided after the mid-term via BrightSpace. In cases of unequal contribution to the group assignment the grade may be individually adjusted accordingly.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible.	
Special Information	Contact: rm-io@tudelft.nl	

IOB6-E3	Health	5
Course Coordinator	A. Albayrak	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Course Contents	Health is an elective course where the students will gain knowledge on designing products, services and interventions in the context of health with its specific stakeholders and conditions. The course will provide an overview of relevant topics, developments and challenges in health, such as advanced health technologies, patient experience, digitalization, regulations, etc. The Dutch health care landscape will be explored by capita selecta lectures and workshops. The lectures and workshops will offer the students different perspectives in the complex health system and will be provided by different stakeholders active in the domain of health, such as patients, informal care givers, healthcare professionals, designers, researcher, industry, etc. This course will also present a sneak preview of the topics, projects and research themes of the Medisign specialization. Medisign is a master specialization at the Faculty of Industrial Design Engineering and aims to educate dedicated and skilled industrial design engineers in the field of healthcare.	
Study Goals	Upon finishing the course students are able to: 1. Illustrate the complexity of the health system and its stakeholders by mapping a patient journey (deepening LO 3.2.1) 2. Understand the different perspectives of the stakeholders and reflect on their own perspectives (deepening LO 5.1.1) 3. Reflect and motivate their own role as a designer in the field of healthcare (deepening LO 5.3.2)	
Education Method	The course consists of guest lectures and workshops. The lectures and workshops will be provided by the different stakeholders active in the domain of healthcare. Each week 4 hours are scheduled for these lectures & workshops. The students will read relevant key publications (both scientific & non-scientific) and prepare questions for the discussion with the stakeholders. The student will choose a buddy in the first week and work with this buddy throughout the whole elective. The buddy is meant to support the process of critical thinking, discussion and feedback. Together they will prepare a patient journey where they map the different stakeholders they have met during the lectures. The reflection report at the end is individual but buddies will review each others draft and provide feedback before handing in the final version.	
Literature and Study Materials	Key publications and/or relevant reports per topic will be offered on Brightspace.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of the course are assessed through an individual reflection report, including a patient journey mapping and the preparation for the discussions (summative). The report will be 100% of the final grade. Students will receive formative feedback by peer review of the pre-reflection paper.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible	
Special Information	Contact: health-io@tudelft.nl	

IOB6-E4-22	Identity and Portfolio	5
Course Coordinator	Dr.ir. M.H. Sonneveld	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The applied goal of this course is to support students in creating a meaningful portfolio that will allow them to present themselves as designers, for example to future coaches, potential companies for internship, graduation projects, possible jobs and so on. To achieve this goal, students will explore their identity as a designer. First, by reflecting on the design experiences they had so far (and collected in the IDME activities). Next, by exploring what motivates them, what inspires them, what values and future vision they have, what their position in the Design Field could be, and so on. In addition, students will explore different means of visual and verbal presentation, to discover which modes of expression fit their identity. This will allow them to align content and form of the portfolio. The coherence of the portfolio will be ensured by emphasizing during the course on the art of storytelling. The course will not conclude with a finished portfolio as a means in itself, but with the foundation for creating professional portfolios in the future. The second goal of this course is to introduce the students to the more theoretical aspects of the different perspectives on the concept of Identity. Students will explore these concepts while creating their own. This theoretical foundation will be valuable for students, first, to have a solid basis for their portfolio, and next, in their future design work, where designing for others always implicates awareness of identity, and the dilemmas this concept elicits.	
Study Goals	Upon finishing this course students are able to: 1. Present themselves, their design work and their future aspirations in an effective, inspiring and convincing way, using the art of storytelling (deepening LO 5.4.3). 2. Position themselves in the field of Design (deepening LO 5.3.2). 3. Understand the relation between content, form and goal of a presentation about themselves (deepening LO 6.3.1). 4. Distinguish between the different aspects and theories on Identity 5. Construct a personal concept of Identity that supports them in their personal presentations	
Education Method	The course offers a blended learning format of online materials, self-study and coach meetings. Students will work individually in studios, assisted by one coach, supported by IDE alumni (2 to 4 per studio). Week 1-4 have the same structure: The week starts with a weekly question for the student to explore individually (self-study), using the materials offered by the course, and by looking for additional resources if desired. This exploration (regarding content AND form of the portfolio) ends with the formulation of the insights gained that week, and the implications of these insights for the building of the portfolio. Each week ends with a coach meeting in groups, where the individual work is discussed, concluding in a group discussion on what has been learned. Week 5 is dedicated to synthesis, final explorations, and formulating the goal of the portfolio students are working on. The students work individually, supported by peer-feedback and formative feedback of the coaches and alumni. At the end of week 6, students collect the insights from week 1 to 6 in a preliminary report, and hand it in for formative assessment at the end of the week. Week 6-9 are dedicated to the building of the portfolio, with the support of peer- and coach feedback. The portfolio is handed in at the end of week 9. The portfolio will be summative assessed by the coaches. Week 10 is dedicated to the final presentations: students will prepare AND present their work to an audience consisting of coaches, students and alumni, who will give feed-back from the perspective of the goal the student set out for. Students will write a final reflection on the learnings in the course. The insights report (week 5) completed with the reflection will be handed in at the end of the week, and summative assessed by the coaches. The feedback of peers and alumni will be used by the coaches to finalize their summative assessment of the portfolio and the presentation.	
Literature and Study Materials	During the course, students will be provided with: - A course manual (according to the provided template) - An introduction video with the goal and structure of the course, and with examples - Background information on different perspectives on Identity (papers, book chapters, and so on) - Instruction videos on modes of expression (developed by the core team in collaboration with experts) - Blueprints for the different assignments - A rubric for the assessment of the deliverables In addition, students are encouraged to individually gather additional information on the specific topics they are interested in.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of the course are assessed through two deliverables: 1. The portfolio the students have created and presented (50%) 2. A report on the insights and implications gained in the first 6 weeks of the course completed with a final reflection (50%) Both will be assessed by means of a rubric.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible	
Special Information	Contact: identity-io@tudelft.nl	

IOB6-E5	Entrepreneurial Innovation by Design	5
Course Coordinator	Prof.dr.ir. F.E.H.M. Smulders	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Course Contents	Dit vak wordt in het NL gegeven, maar alle literatuur zal in het Engels zijn. This course focuses on the organizational and behavioral context of the innovative design process, its relationship to startups, to socio-technical systems and to entrepreneurial behavior. At IDE we train designers to develop new and innovative products/services/systems from initiation to validated concept, and no further, so no implementation! The course will extend this 'limited' focus in two ways: first, to encompass the trajectory from concept to realization; secondly, to include the perspective of innovation actors other than the designers themselves (stakeholders). In the course you will learn that the organizational actors from different disciplines should be seen as the first users of the concept delivered by them/designers. The course will focus on these so-called 'intermediate users' who sit between the design concept and the end user, which is your traditional focus. You will learn to assess the organizational and behavioral context of the intermediate users with a view to designing (innovative) concepts that are ready for implementation. In addition, you will learn that the implementation of outside-the-box concepts requires entrepreneurial behavior on the part of most organizational actors to come to fruition. To ensure that the concepts from theory become relevant, this course has been set up as a Mixed-Classroom course in which students work together with an innovation professional from the field. From week 5, each student team has its own customer with an organizational innovation challenge. As a team, you will be asked to assess the organizational context, understand the challenge and provide a workable response. Most customers will speak Dutch. Finally, you will also reflect on their individual personality traits from the perspective of the entrepreneurial journey from concept to realization.	
Study Goals	Upon finishing this course, students are able to: 1. Apply theoretical framework of implementation/scaling processes to design concepts for the purpose of realization planning (deepening LO 1.1.3) 2. Assess and appraise organizational processes with the purpose of mapping absorptive and innovative capacities in relation to concepts 3. Explicate and contrast the theoretical principles of designing, engineering, innovating and entrepreneurship 4. Describe differences and similarities between new-venture and corporate entrepreneurship 5. Know how to apply the notions of frontloading, disciplines and intermediate users to the case study 6. Apply their understanding of technology development from the perspective of socio-technical systems to the situation of concept design 7. Explain theory by a lecture to inform an audience of lay people (deepening LO 6.3.2) 8. Show awareness of personality traits and apply self-knowledge for envisioning their role within entrepreneurial innovation processes (based on personality tests) (deepening LO 6.2.2).	
Education Method	The course consists of the following education methods: 1) Online modules 2) Individual assignment 3) Group assignments + Fishbowl sessions (wk 2, 3, 4, 5) 4) Group assignment + Conference setting (wk 6 & 10) Ad 1) Series of online modules run in parallel to the two group assignments Ad 2) Personality tests will assist students in creating awareness on their given qualities in relation to innovation activities Ad 3) Literature will be studied, presented and discussed in Fishbowl sessions. The final deliverable will be a team/personal glossary on entrepreneurial innovation. Ad 4) Real life case study brought by practitioners.	
Literature and Study Materials	Online, Video's, tutorials & Papers, announced via Brightspace.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industriel Ontwerpen).	
Assessment	Assessment of individual work, team reports and team lecturing. The final grade consists of the individual assignment (20%) and the group assignment (80%; including lecturing and project).	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible	
Special Information	Contact: eidos-io@tudelft.nl	

IOB6-E6	Design Engineering Contest	5
Course Coordinator	Dr.ir. A.J. Jansen	
Contact Hours / Week x/x/x/x	3,5	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This course allows students to experience the day-to-day reality of engineering design; team work, deal with competition, scarce resources and limited time. During this course students (in groups of max. 5) will design and build a robot that has to perform what seems to be a simple task. Every group will start from an identical toolkit containing electric motors, gearboxes and a variety of mechanical parts. An exciting competition on Saturday April 1 (no joke) will determine who can solve the task best while dealing with the competition at the same time. And like in the real world, anything could happen during the runtime of the course. Learning objectives of this course include team work, adaptation skills, practical electronics and prototyping during a design-build-test cycle. During the course, students will build upon team working skills, by reviewing their team performance using the 7C model (Janssen, 1999). The two deliverables of the course are a report on the robots performance and an individual journal in which students reflect on their role as a team member.	
Study Goals	Upon finishing this course students are able to: - Design and build a robot that can perform a simple task (deepening LO 1.6.1, LO 2.5.1, LO 2.5.2) - Contribute to team work effectively and reflect on their contribution to the teams performance (deepening LO 5.1.2, LO 5.1.3, LO 6.2.1 & LO 6.2.3). - Analyze and estimate the success rate of their own robot versus the competitors robot and act upon this (deepening LO 1.5.3, LO 1.6.1 & LO 1.6.2)	
Education Method	The education method of this course is best described by using the format of Challenge based learning (CBL) in which students mainly learn by doing (design, build, test cycle). Student groups have to be self-propelled in acquiring the appropriate skills and knowledge to complete the task at hand. There is significant room for self-reflection. Student coaches will guide both the learning process and content. Lectures are mainly aiming at structuring the learning process by pointing the students at useful knowledge sources, not purely at transferring knowledge. The course will require students to become aware of their behaviour in teams and reflect upon this in a methodical way at regular intervals.	
Literature and Study Materials	Materials will be announced on Brightspace.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The final grade is based on two deliverables: 1) self-reflection journal (individual, 60% of the final grade) 2) Final report containing a product performance analysis (group work, 40% of the final grade). In this grade also the success of the robot in the contest will be taken into account.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. MIND: the contest is scheduled for SATURDAY April 1st (no joke) so keep this day available in your agenda! Registration beyond the registration deadlines is not possible	
Special Information	Contact: dec-io@tudelft.nl	

IOB6-E7	Biomechanics of Product Use	5
Course Coordinator	Prof.dr.ir. R.H.M. Goossens	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	In this elective, students will learn to conduct simple biomechanical analyses of human-product interaction. Furthermore, they will learn about how to model, analyze, and evaluate musculoskeletal system of human beings during daily activities and product use. It gives an introduction in the different structures in the body (like bones, ligaments, and tendons), their mechanical properties, and the mechanical load in order to develop design requirements.	
Study Goals	Upon finishing this course students are able to: 1. Apply basic anatomy, mathematics and physics in the context of biomechanics in product use. 2. Systematically model, analyze, and evaluate the biomechanical behavior in simple static and dynamic situations (deepening LO 2.1.2). 3. Apply biomechanical principles aligned with the human body (including bodily functioning and psychophysics) to create design requirements (deepening LO 2.6.1).	
Education Method	The course consists of lectures and/or Q&A, and a project assignment. The lectures and/or Q&A are held at least once a week, and provide the student with information on how to execute the project assignment. The project assignment is done individually, and consists of making simulation of a biomechanical application in OpenSim, and writing a report.	
Literature and Study Materials	Mandatory: 1. Nordin, M., & Frankel, V. H. (Eds.). (2001). Basic biomechanics of the musculoskeletal system. Lippincott Williams & Wilkins. 2. Manual: Self-instruction assignment Biomechanics. 3. OpenSim Users Guide	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of the course are assessed through a individual report (summative assessment). Formative feedback is provided during interactive Q&As.	
Enrolment / Application	For IDE BSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which the app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.	
Special Information	Registration beyond the registration deadlines is not possible Contact: bfpu-io@tudelft.nl	

IOB6-E8	Design Analytics	5
Course Coordinator	Dr. H. Verma	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this technological and methodological elective, students will gather tools and acquire knowledge to critically evaluate (interactive and intelligent) technological prototypes and set the stage for the next iteration of the design phase by eliciting actionable design guidelines based on comprehensive empirical analysis. The technological component of this course is demonstrated by the strong focus on the improvement of IPSS (intelligent products, services, and systems) such as, conversational assistants, wearables, etc. Instead of examining the complete design cycle, students will focus solely on the quantitative assessment of IPSS and examine their cognitive, perceptual, and experiential effects. To this end, students will learn and add skills to their methodological toolkit, which will enable them to a) design sound experiments (with Control and Experimental groups), b) use sensing technologies (e.g., eye-tracking) and adapt existing data-acquisition tools (e.g., NASA TLX, MSLQ), c) identify the right statistical tests (descriptive vs. inferential; comparison vs. regression), d) discriminate correlation from causation and identify confounding factors, and e) understand the proper means of communicating their quantitative analysis and results. The content of this course is relevant for students because they learn how to systematically assess their designs with standardized practices and communicate their findings with multiple stakeholders and a broader design-research community. In this way, this course offers an essential ingredient in students transformation from a UX Designer to a UX Researcher. In this course, the student will become familiar with the 5-stage Design Analytics Process (DAP) i.e., design diagnostics, study design synthesis, experimentation, data analysis, and design prescription.	
Study Goals	Upon finishing this course, students are able to: 1: Describe the design unknowns or flaws of a given product, service, or system (deepening LO 1.3.3, LO 1.6.3). 2: Formulate research questions and pertinent hypotheses to empirically evaluate the given technology (deepening LO 4.2.1). 3: Design an experiment protocol to validate the research hypotheses (deepening LO 4.2.2, LO 1.6.1). 4a: Evaluate the quality of collected multi-modal data (deepening LO 2.4.4). 4b: Apply the pre-designed experiment protocol with real subjects (deepening LO 1.6.1) 5: Analyze experimental data using appropriate statistical methods (deepening LO 2.1.1). 6: Explain the results of statistical analysis for future design improvement (deepening LO 4.3.1, LO 4.3.2).	
Education Method	This 10-week course consists of 9 weekly lectures (2h/week), 9 weekly hands-on workshops (upto 1.5h/week) with Teaching Assistants, and 9 weekly coaching sessions (0.5h/week). Students are expected to prepare themselves for an upcoming lecture by studying the provided reading material and by watching the provided educational videos in a flipped classroom format. Lectures and hands-on-sessions will be related in terms of content to ensure that the theory--in the lectures--will be put into practice--in the hands-on workshops. Moreover, student groups will have weekly discussions with coaches, who will guide them in their particular design briefs. Inherently, this course provides the ideal opportunity for applying an inquiry-based learning approach at scale. As such, we will invite colleagues, interested parties, and stakeholders to submit their IPSS (Intelligent Products, Services, and Systems) designs for evaluation a few months before the start of the course. Group assignments At the end of the course, the student groups are expected to submit a comprehensive report documenting their engagement with the 5-stage Design Analytics Process, and their empirical evaluation of the chosen IPSS. Weekly sessions with the coach (formative assessments) will allow groups the possibility of seeking feedback on their project direction and progress. In week 9, student groups will offer consultation sessions, in the form of a presentation followed by panel discussion, to their respective clients explicating the identified design guidelines and potential routes for design improvements. Kickstarting lecture and hands-on session (week 1) At the kickstarting lecture, we will employ ice-breaking techniques (e.g., "speed-dating") for enabling students to familiarize among themselves. Next, we will perform a brief diagnostic assessment by using the Kahoot! game-based learning platform. Students will interactively answer questions that determine their personality profiles (e.g., the Big-5), interests, and grouping preferences so that we form effective and diverse student teams (3--4 persons). After a brief presentation of all the available IPSS designs, the newly formed teams will bid for the design they prefer to evaluate throughout this course by delivering a final report at the end of the course. Intermediate lectures and hands-on sessions (weeks 2--8) In each lecture, we will showcase the research and experimental thinking that corresponds to the given stage in the course, and in full alignment with the established LOs. To this end, we will present examples of existing IPSS designs tested through our 5-stage DAP, outlined above, in the form of "evaluation cases": An evaluation case can be a prototype borrowed from literature where we describe how specific design unknowns are targeted for formulating concrete RQs/Hypotheses. During lectures, we will define our threshold concepts--notions that once understood change the perception of a subject, phenomenon, or experience. During the lectures, we will use Kahoot! for increasing engagement with brief quizzes and for collecting learning feedback in situ. In the following hands-on sessions, the comprehension of the threshold concepts will be tested with practical work on other evaluation cases. Then, the students will plan how they will move forward with their evaluation cases in collaboration with coaches. Final Lecture (week 9) The last lecture is dedicated to the presentation of the group work. Each group of students will present their findings publicly to an audience composed of the stakeholders, who supplied our course with their IPSS designs, other students, and interested parties. A panel discussion will follow each presentation where each student team answers questions from the audience.	
Literature and Study Materials	The course coordinators will design a reader on Brightspace with a curated set of readings, lecture notes, web resources, hands-on tutorials, and videos for different modules within this elective. The mandatory weekly readings will be selected chapters from a list of relevant books, and other articles.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	The learning objectives of this elective will be assessed through a group report, including a final presentation, and an individual exam. Group deliverables will contribute to 60% of the students final grade. Students will weekly self-assess their contributions,	

as well as of their peers in an anonymous manner. These self- and peer-assessments will contribute to a proportion of their grade for the group work. The remaining grades will be assigned by respective coaches. The final exam will contribute to the remaining 40%.

Group Work (60%): Student groups will deliver a final report at the end of the elective (max. 20 pages), accompanied by a final presentation. This group deliverable corresponds to the aforementioned DAP stages and the underlying LOs. The LOs are fulfilled by the rigorous evaluation of an IPSS (design artifact), for which the groups will work throughout the duration of the course. The course coordinators will provide a template and assessment rubric for these reports, and instructions on the length and structure of each report. The final report will manifest into a research portfolio that will be presented publicly at the end of the course to an audience composed of the stakeholders, who supplied our course with their IPSS designs, other students, and interested parties. The final individual grade for the group work will be the sum of the grades for each group report, weighted by the peer contribution assessment scores produced weekly.

Peer Contribution Assessment: Every week, students will assess the contribution of their peers as well as their own. The coaches will then produce individual student grades for the final deliverable by weighing the average peer contribution assessment score throughout the course.

Individual Exam (40%): A final exam will constitute a mix of MCQs (multiple choice questions) and open questions.

Formative Assessment: Weekly group coaching will provide students with the necessary feedback as well as enable them to assess the progress of their learning, their conceptual understanding. In addition, group coaching will assist students to effectively align their project direction by reflecting critically about the diverse aspects of design analytics.

Enrolment / Application

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Registration periods and deadlines are communicated on the website. See <http://www.io.tudelft.nl/osiris> for exact deadlines and details.

Registration beyond the registration deadlines is not possible

Special Information

Contact: da-io@tudelft.nl

IOB6-E9-22	Form and Senses	5
Course Coordinator	Dr. E. Ozcan Vieira	
Contact Hours / Week x/x/x/x	3	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Form & Senses is a multi-sensory design course in which students gain knowledge to understand and explain sensory perception of products and skills to manipulate sensory product properties to be able to design targeted sensory experiences (e.g., warm sounds, calm textures, cosy lights). Students in particular develop their knowledge and skills by practicing sound design, texture design and lighting design. Furthermore, students are given a chance to make use of their previous knowledge and skills. In a group of four/five, students are encouraged to practice their existing skills in using (digital) technology to build multisensory sketches, to ensure sensory integration, to envision and explain targeted experiences, and to research whether the envisioned sensory experiences take place. In the first half of the course students explore the sensory design space with mini-exercises focusing on multisensory form transformation (e.g., how does a plant sound like? What texture does this sound have?). As an assignment, students will explore a multisensory design space by creating a multidimensional and interactive assemblage that has been validated to evoke desired experiences (e.g., warm sounds, textures, and lighting). Form-S deepens students knowledge and skills for form giving and experience design. Students are given a chance to explore the conceptual design for sensory product properties.	
Study Goals	Upon finishing this course, students are able to: LO-1. analyse and apply the principles of sensory product form (e.g., sound, vision, tactility) in their multi-sensory concepts (i.e. sensory sketching) and design vision. (deepening LO 2.7.2) (deepening LO 2.6.2) LO-2. analyse and argue the impact of sensory form on human perception and experience. (deepening LO 2.7.2) LO-3. create a multisensory object and an interactive and multi-dimensional assemblage (e.g., collage, display, space, canvas) that evokes a desired experience. (deepening LO 2.7.2) (deepening LO 2.7.1) LO-4. assess the experiential quality of their multisensory object and assemblage and communicate their findings verbally and in text. (deepening LO 2.6.2)	
Education Method	Mini theoretical lectures (15 min) and scientific papers are provided on Brightspace for self-study (Weeks 1-5). Expert and artist lectures in a lecture hall are reserved for discussions on dedicated sensory experiences (Weeks 2-4). Students will explore the elements of multisensory design in the form of sensory sketches (Weeks 2-4). In week 5, students receive formative feedback on the sensory quality of their sketches. Students will have one major assignment that integrates sound, texture, and lights in one congruent multisensory design space (Weeks 5-9). Students will also work in groups of four/five students allowing for discussions and idea exchanges (Weeks 2-10) under the supervision of a design coach. The assignment will have both an individual (i.e., reflection and essay) and a group component (i.e., the design space (objects and assemblage) and experience card). Technology and Sensory Experts will be available on request to provide relief for possible set-backs (Weeks 5-9). Students will finally demonstrate their process for their multisensory design space with the help of a experience card.	
Literature and Study Materials	Scientific papers will be used in the course. All study materials (scientific papers, online lectures, etc.) will be shared on the Brightspace. A course manual (shared on Brightspace) will guide the students throughout the course on the planning and deliverables.	
Prerequisites	Only students can be admitted who, after completing the selection procedure, were selected for the Industrial Design bachelor programme at the Delft University of Technology (article 5 - OER Bachelor Industrieel Ontwerpen).	
Assessment	Students will be assessed by the quality of the group project that entails creating multisensory objects and documenting the process and by the quality of the individual essay. Students will be formatively and individually assessed to provide feedback in Week 5 so that the input on personal development towards multisensory design could be used for the final summative assessment of the group projects. The input from the formative assessment also helps guide students towards the summative assessment of the Individual Vision Essay. Deliverables for Assessments: - Midterm Deliverables (Individual) - a multi-sensory object as the result of the design exercises (Week 5 Formative) - sensory experience card(s) that document the multi-sensory design process (Week 5 Formative) - End-term Project Deliverables (Group) - Interactive and Multi-sensory Assemblage demonstrating the ability to think and create for multi-sensory concepts using their existing and new knowledge and skills (Week 9 Summative) - Multi-sensory Experience card on the creation of the assemblage between Week 5-9 (Week 9 - Summative) - End-term Individual Deliverable - Vision Essay on the Assemblage (Week 10 - Summative) - Summative Group Assessment (Project) - Grade - Multi-sensory Assemblage (Week 9) - Allows for team discussion and decision making on how to apply principles of sensory product form (LO-1; LO-2; LO-4) - Expresses interactivity and elicits sensory experiences (LO-3) - Allows for experience measurement (LO-4) - Experience cards (Week 9) - Argue the varying effects of multisensory experiences on human product perception (LO-2). - Use relevant vocabulary for explaining different sensory experiences (LO-4) - Summative Individual Assessment (Essay) - Pass/Fail - Vision Essay (LO-2) in Week 10 Final grade = 100% - Group grade (project) AND Pass/Fail - Individual grade (essay) In order to pass the course, students will need at least a 6 as a grade for their group work and pass the essay assignment.	
Enrolment / Application	Registration in MyTUDelft is required for all IDE courses (article 13 - OER Bachelor Industrieel Ontwerpen). Enrolment is possible during the registration period which closes about 5 calendar weeks prior to the first day of the semester in which the	

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program starts. The faculty announces the exact deadline for registration timely. The registration for courses has to take place per semester, meaning for courses of 2 quarters at once.
See <https://www.tudelft.nl/studenten/io/praktische-zaken/aanmelden-voor-vakken> for deadlines and details.

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